

No Longer Business as Usual: Evidence from Two Literacy Interventions in Ghana

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Almost all evidence on education programs comes from settings where the counterfactual is business-as-usual schooling. As effective instruction scales, the policy question becomes what a program adds on top. I run a randomized trial in 103 Ghanaian schools already implementing an effective structured pedagogy program, testing two additional sources of learning: parents and principals. Schools received (i) restructured parent meetings plus take-home copies of the literacy workbook used in class, (ii) instructional leadership training for principals, (iii) both, or (iv) neither. The parent program raises reading fluency by 0.33 standard deviations, while the principal training lowers reading comprehension by 0.26 standard deviations. Combined, the parent program offsets part of the comprehension loss caused by the principal program. I show this is consistent with concave literacy production: the marginal return for parent engagement increases when instruction quality drops, so families invest more time in at-home learning.

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The literacy gap between rich and poor countries remains persistently large despite recent advances in improving classroom instruction.¹ Proven literacy programs can deliver average learning gains of 0.85 standard deviations when implemented well (Angrist and Meager, 2023) which, while impressive, closes less than half of the literacy gap between Sub-Saharan Africa and rich countries (Angrist et al., 2021). Policymakers have piloted and scaled literacy programs targeting learning constraints through many channels. Programs that improve teaching by providing high-quality materials or targeting instruction create the biggest learning gains but are intensive for school systems to implement. Other promising programs activate parental investment in school or early childhood, treat health conditions at school, or ease school access to improve learning with a lighter touch (Akyeampong et al., 2023). Alone, even the best of these programs will not be enough. Yet almost all programs are tested against a counterfactual of business-as-usual schooling.

That counterfactual is disappearing. Across the developing world, effective instructional programs are scaling quickly. In Botswana, every government school now uses targeted instruction (Angrist, Matsheng, and Perlman, 2026), while a structured pedagogy program has been running in 22,000 Kenyan schools since 2017 (Piper et al., 2018a).² The policy question is no longer whether a program beats the status quo, but what it adds to a system that can already teach well. This distinction matters because learning gains from an add-on program need not resemble its returns when tested alone. If strong classroom instruction has already relaxed the binding constraint on learning, further programs may do little, but if programs target different constraints, learning gains should persist.

In this study, I estimate learning gains from add-on programs directly using a randomized trial with a factorial design in the Central Region of Ghana across 103 public and private schools. All of these schools were already receiving a highly effective structured pedagogy program (Andersen et al., 2026b), so my estimates identify each program’s effect as an addition to strong classroom instruction. I study the individual impact of two programs designed to enhance literacy through channels outside the classroom, as well as their interaction. The first program aims to increase parental involvement by redesigning existing parent-teacher

¹ For an overview of effective education programs, see (Angrist et al., 2025).

² In Ghana, a differentiated learning program reached 2.4 million students in over 10,000 schools in 2026.

meetings. Parents practice reading with their children and are sent home with the literacy workbook used during literacy classes to facilitate further reading and practice. The second program trains principals in instructional leadership through a two-session in-person, video-based course covering classroom observation and coaching of literacy teachers, and the effective use of assessment data. I randomly assigned 26 schools to each treatment arm, yielding a sample of 2,089 first-grade students.³ The study ran during the 2025/2026 school year with both programs implemented at the end of the first term.

The parent program causes a significant increase in reading fluency ($d = 0.33$, $p = 0.04$), my primary outcome.⁴ Treated students read 9.2 words per minute more than the control group average of 30.3, which is a 30% increase in reading speed. They are also 14 percentage points more likely to be able to read at least 45 words per minute ($p = 0.05$), the fluency empirically associated with full reading comprehension (Abadzi, 2012). I also find positive but imprecise effects on reading comprehension ($d = 0.16$, $p = 0.22$).⁵ These gains are particularly striking because control group students already read nearly five times faster than the national average (Service, 2015), and the fluency effect puts the parent program at the 75th percentile of all literacy programs (Evans and Yuan, 2022).

In contrast, the principal program reduces literacy. The principal program reduces reading comprehension by 0.26 SDs ($p = 0.05$), with a smaller though noisily estimated decrease in fluency (0.11 SDs, $p = 0.53$). This is a large decrease in reading: it is below the 10th percentile of literacy programs in low-income countries (Evans and Yuan, 2022). Ten percentage points more children cannot answer a single reading comprehension question correctly ($p = 0.06$). The lack of gains is not due to implementation failure: 88% of assigned schools attended at least one of the course sessions, and attendees on average scored 94.9% on a short knowledge quiz. Nor is the decrease an artifact of an unlucky randomization draw. If any-

³ 103 schools is not evenly divisible into four groups, so the parent program-only arm had 25 schools.

⁴ My pre-analysis plan specified that my primary outcome would be letter sound knowledge. However, only 30% of assessments included this EGRA module due to an issue with different versions of the app teachers used to run the assessments. In lieu of letter sound knowledge, I use oral reading fluency because it is the most foundational reading skill I test. Appendix B presents treatment effects on letter sounds.

⁵ All p-values are from two-sided randomization-inference hypothesis tests, which I report throughout as a conservative default. My pre-analysis plan specified positive, one-sided hypothesis tests; under those, reading fluency remains significant ($p = 0.02$) and comprehension does not ($p = 0.12$). For the principal program, the negative point estimates make the registered one-sided tests uninformative, and two-sided reporting follows the plan's provision for negative effects.

thing, before treatment, principal arm students had slightly higher reading comprehension.⁶

My factorial design also lets me analyze if the two programs interact. The parent program offsets the reading comprehension loss the principal program causes to some degree. When combined, the programs produce an effect on reading comprehension 0.2 SDs larger than the sum of their individual effects. Although I cannot statistically distinguish this interaction from 0, I bound the interaction between -0.11 and 0.5 SDs. A positive interaction implies the programs are substitutes in the production of literacy or equivalently, the parent program had a larger effect in schools also receiving the principal program. I show suggestive evidence that parents increase their engagement when school quality drops, and this effect is more pronounced when parents are empowered through the parent program. Parental engagement increases by 0.39 SDs ($p = 0.24$) in the principal arm and by 0.44 SDs ($p = 0.22$) in schools receiving both programs. Specifically, in schools receiving both programs, parents seek out and discuss reading test results with principals 44% more often ($p = 0.01$). This pattern is consistent with strongly concave production of literacy. When the principal program reduces instruction quality, this increases the marginal returns to home investments in learning, so parents invest more, which partially offsets the losses.

Several pieces of evidence indicate the parent program’s primary mechanism is supplying practice material rather than changing parents’ behaviors. First, the program raised the proportion of households with the literacy workbook at home by 15 percentage points. The meetings themselves were not the only vehicle for workbook dissemination: possession is identical for parents who attended meetings and those who did not (82% vs 80%). In low-income countries, parents can lack information on how well their children can read and so purchase books that are mismatched to the children’s ability level (Dizon-Ross, 2019); the literacy workbooks are by construction matched to the material students learn in class. Parents respond to having the materials by buying 10 percentage points fewer other books. Household behavior, by contrast, cannot fully explain the learning gains. Parent-reported engagement does not move ($p = 0.92$); teachers report a 47% increase in one-on-one meetings with parents ($p = 0.05$). Parents, however, are not the only actors who support children’s

⁶ Principal trainings affect all grades within a school, but my data only covers first-grade students. My data would be blind to changes in literacy in other grades, and so may understate the school-wide effect of the program.

learning at home ([Jakiela et al., 2025](#)): the parent program decreases the number of students who have sibling-supported learning by 25% ($p = 0.17$). To account for this, I survey students’ at-home learning engagement by any family member, not just their parents. This measure increases by 0.56 SDs ($p = 0.09$) due to the parent program. Dissemination of the literacy workbooks is a plausible channel through which other family members who did not attend the meetings could benefit from the program and drive the large learning effects. I find no heterogeneity for the program’s effects across gender, school type, teacher experience, or parents’ education. These facts are consistent with appropriately leveled workbooks driving the effect by increasing the marginal productivity of time spent reading, no matter who, if anyone, assists.

The principal program produced reduced literacy despite evidence of strong implementation. At their schools, principals logged 62% more lesson observations and coaching sessions of their first-grade teachers on their school management app. However, this uptick is not verified by my survey data. Self-reported observation frequency by both principals and first-grade teachers do not change. Part of the training promoted app use for principals, so this discrepancy could be explained by treatment shifting reporting rather than behavior. Despite no change in observation quantity, I show teacher behavior changed. Teachers were more rigid in sticking to lesson plans ($p = 0.19$) and 26% less likely to frequently grade student work ($p = 0.01$). At the same time, teachers report feeling more stressed by principal feedback ($p = 0.08$). While fewer deviations from high-quality lesson plans may sound beneficial, experienced teachers are often skilled at making small deviations that better target the instruction level to their students ([Jackson and Makarin, 2018](#)). Strict adherence to lesson scripts pushed by principals may replace expertise and harm student learning. I attribute these effects to a shift in the content of lesson observations driven by treatment. While the quantity of observations did not change, they may have become more formal. When the observation is recorded on the administrative app, it produces a quantifiable score “grading” the teacher on their lessons. This may have caused observations to shift towards a more formal structure that put more pressure on teachers, causing the increase in stress.

My results contribute to three literatures. First, I provide experimental estimates of the returns to education programs against a counterfactual of strong classroom instruction

rather than business-as-usual schooling. Most experimental evidence on education programs in low-income countries comes from studies in untreated systems, but as effective programs scale, the relevant question is what a program adds to them. This philosophy mirrors a movement in program design to use A/B tests to refine programs against their own success rather than the status quo (Angrist, Cullen, and Magat, 2025; Andersen et al., 2026b). I extend this philosophy by treating structured pedagogy as the relevant counterfactual to building successful education programs and show that parent programs can achieve large learning gains on top of already high learning levels. My results are also part of a literature on factorial experiments. Interactions in education production are of first-order importance (Glewwe and Muralidharan, 2016). Some program components build on each other (Mbiti et al., 2019; Kerwin and Thornton, 2021) while others undermine each other (Wolf et al., 2019). Factorial experiments estimate this degree of complementarity (if any) rather than only whether effects stack (Attanasio et al., 2014; Nyqvist and Guariso, 2026). I add among the first factorial evidence combining home- and school-based programs, and estimate their interaction. I show that behavioral effects are a key driver of interaction effects, driven by evidence that literacy production is concave in home investment.

Second, I contribute to the literature on parental engagement in education. Parents have long been considered crucial to education production (Todd and Wolpin, 2003), but parental involvement has proven hard to move in practice (Di Maro et al., 2024). I distinguish three channels through which parent programs can increase learning. The first reduces the cost of education investments. Most parent programs target this channel. Behavioral nudges through text messages can make education more salient and increase investment (Doss et al., 2019; Angrist, Bergman, and Matsheng, 2022). This can work at scale (Goyal et al., 2024). Parents, however, may lack information on their children’s ability level and make sub-optimal investments (Dizon-Ross, 2019). Providing that information can correct parents’ misperceptions and increase children’s learning (Andrabi, Das, and Khwaja, 2017; Barrera-Orsorio et al., 2020; Bergman and Chohanajin, 2026; Berlinski et al., 2025). But cheaper investment is not always enough. Parents may lack the capacity and organizational ability to act on nudges or information (Yedomiffi, 2025). The second channel builds that capacity through maternal literacy lessons (Banerji, Berry, and Shotland, 2017) or manage-

ment training (Yedomiffi, 2025). These programs can be demanding for families, so a third channel raises the productivity of time spent with children by supplying materials (Knauer et al., 2020; Mo et al., 2014). This approach has produced mixed results. I contribute to these existing findings by showing why this third channel can work. Leveled workbooks make any time spent reading more productive, so learning gains do not depend on parents investing more or acquiring new skills. The scripted practice material makes up for knowledge parents may lack, which is the same active ingredient of the most effective classroom-based literacy programs (Piper et al., 2018c). I show this approach can successfully create large learning gains at home and that these gains concentrate in higher-order skills that many literacy interventions do not shift.

Third, I provide evidence on when school management training is counterproductive. The quality of management strongly predicts school performance (Bloom et al., 2015) but interventions to improve it have produced mixed results (Popova et al., 2022). A meta-analysis found that management reforms produce only a 0.033 standard deviation improvement in student learning (Anand et al., 2023). Training principals nonetheless remains an attractive, low-cost lever because training one person can benefit an entire school. One prominent reason is the long causal chain from management to student learning. Intermediate administrative outcomes can improve without translating to learning (Muralidharan and Singh, 2020), and training often fails to change teacher practice at all (Loyalka et al., 2019). I build on this evidence by showing that failure can occur at the final link of the chain. The training reached teachers and changed their behavior, yet literacy fell despite every intermediate outcome pointing to program success. I show that intermediate outcomes do not serve as cheap proxies for learning; they can point in the wrong direction and give false positive signals.

The remainder of the paper proceeds as follows. In [Section 1](#), I describe the setting of the study and both interventions. I describe the data I use in [Section 2](#) and discuss program take-up. [Section 3](#) describes the empirical strategy I use to analyze the data, and [Section 4](#) implements that strategy and presents results. [Section 5](#) concludes.

1 Setting & Intervention

1.1 Setting

In Ghana, primary education is compulsory starting at age four with two years of kindergarten and lasts until sixth grade. Compulsory education lasts until the end of three years of junior high school. High school entry is a competitive process that admits up to 150,000 students a year ([Ghana, 2025](#)). 72% of Ghanaian children are enrolled in public schools, but a growing number opt for private instruction. My study covers both. Public schools are free, while private schools can range from low-fee schools with fees less than 40 USD per term to high-end schools that are much more expensive. Teachers in public schools typically have more training than those in private schools, particularly in primary schools ([UNICEF, 2023](#)).

Although overall enrollment in primary school is very high at 98% of eligible students, learning levels are alarmingly low. At the end of second grade, could read only 2.5 words per minute on average, and 77% could not read a single word ([Impact, 2018](#)). The Central Region and Cape Coast specifically, where this study takes place, have among the highest literacy rates in Ghana, but overall literacy is still low ([Service, 2021](#)).

1.2 Intervention

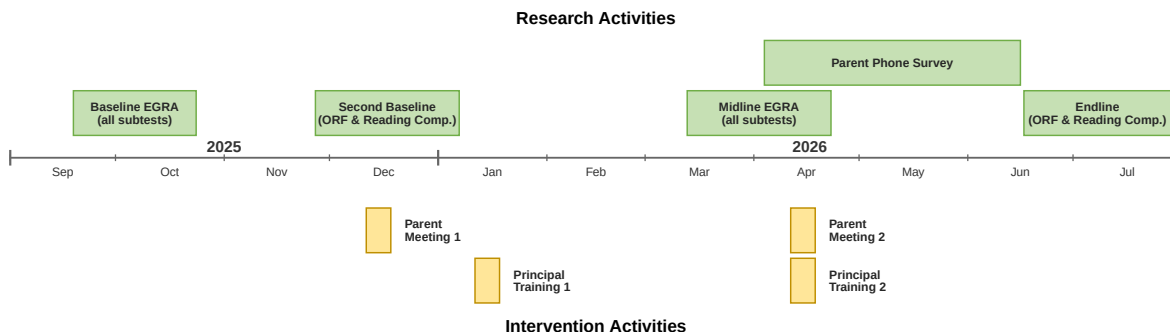
I study two interventions targeting first-grade students, both implemented by Inspiring Teachers, Ghana (ITG). In this section, I provide an overview of the key features of each program. [Appendix F](#) and [Appendix G](#) provide the materials provided as part of each intervention as well as examples of training guides used by the implementers. [Figure 1](#) shows the implementation timeline.

1.2.1 Schooling Baseline

All the schools in my sample were already using an effective literacy program that combines structured pedagogy and targeted instruction.⁷ Tools for Foundational Learning Improve-

⁷ See [Andersen et al. \(2026b\)](#) for a more detailed discussion of the program and its effects.

Figure 1
Study Timeline



ment (TFLI) trains teachers to use “science of reading” principles to build foundational reading skills in students.⁸ Teachers are given structured lesson guides and use these to deliver four one-hour structured literacy lessons a week. On the fifth day, teachers administer a brief literacy assessment of the whole class, then provide targeted reteaching or additional practice based on student performance. This combination has been highly effective so far. TFLI improved reading scores for first-grade students in low-fee private schools in this setting by 0.5 standard deviations on average, with bigger effects for weaker readers. Compared to the status quo, instructional quality in these schools is already at a high base level.

1.2.2 Parent Program

The parent intervention consists of structured parent engagement meetings operating through existing parent-teacher meeting infrastructure. The meetings are facilitated by trained principals and first-grade teachers at each school and run at the end of each of the three academic terms.⁹ Each meeting ran for about an hour during school hours, so some schools had students attend with their parents.

The meetings target three barriers to parental engagement. The first is to convince parents of the benefits of both the science of reading approach used at school and of their participation at home. Parents can be wary of new reading philosophies they were not

⁸ See [Alvarex Martinelli et al. \(2025\)](#) for an overview of the science of reading.

⁹ This includes a meeting run in July 2026 at the end of the academic year. This meeting occurred after I collected reading data, so treatment effectively includes just the first two meetings.

taught in school. That wariness, combined with meetings that empower parents to be more involved in their children’s schooling, can undermine successful education interventions (Wolf et al., 2019). Second, parents receive practice reading materials they can use with their children at home. These materials are the workbook students use during their literacy class. TFLI provides literacy workbooks matched to teachers’ structured lesson guides. The workbooks include places for students to participate in literacy exercises, including copying words teachers write on the board and short stories to practice reading. Some sections include black-and-white pictures of new words. The workbooks also include a literacy report card, which the meetings highlight. Teachers and principals are encouraged to send home the workbooks with students whose parents could not attend the meetings. Third, parents practice readings from the workbook. When children were not present, they practiced with other parents. This includes learning how to run a simple one-minute literacy assessment that mirrors the method used by teachers. The meetings emphasize that parents can help their children practice reading even if they have low literacy. Parents were encouraged to listen to their children read, ask questions about the stories to build comprehension, and discuss the pictures.

ITG staff trained principals and first-grade teachers during two facilitated sessions held in the week before each meeting. Principals and teachers were paid 10 GHS (about 1 USD) to attend each training. Since the literacy workbooks are printed as a regular part of TFLI, this training fee is the only direct marginal cost to run the intervention. At scale, government personnel to run the trainings would be an additional marginal cost, but this would be small per student: each training accommodated principals and teachers from up to 40 schools.

1.2.3 Principal Program

The principal program aimed to boost school quality by training principals in key managerial competencies. Principals attended two one-day in-person trainings held in January and April 2026. ITG trained the principals using a structured series of 18 short videos totaling approximately two hours of content, embedded in a five-hour facilitated session that incorporated guided note-taking, discussion with other participants, reflective exercises, and comprehension quizzes. Because the content is video-based, it is standardized and simple to

scale to more schools.

The training targets two managerial competencies. First, it builds principals’ capacity to coach their teachers. Principals were trained to recognize the key components of the TFLI program so they can effectively evaluate whether their teachers are delivering the curriculum. The training equipped leaders with structured coaching cycles: observe a lesson using a clear focus, provide specific feedback with one actionable step, and guide the teacher in practicing the new technique. Principals practice applying this cycle to realistic scenarios during the training session. The training also covers how to run data-driven progress meetings with teachers using two distinct agendas: reviewing student assessment data to identify struggling learners and plan remediation (assessment to action), or targeting a specific teaching technique for collective improvement (teaching improvement).

Second, the training builds principals’ fluency with the app-based school management system included as part of TFLI, SmartCoach. Principals practice recording classroom observations, tracking student oral reading fluency results, and monitoring implementation through the app. By anchoring coaching in observable data captured through SmartCoach, the intervention aims to shift school leaders from administrative oversight toward instructional leadership focused on measurable learning outcomes.

2 Experiment and Data

2.1 Experimental Design

My data comes from a randomized trial in Ghana’s Central Region, centered around Cape Coast. The treatments targeted first-grade students during the 2025/2026 academic year. The parental engagement treatment arm had its first meetings in late November and early December 2025, a few weeks before the end of Term 1, and the principal treatment arm had its training in January 2026. Both treatments ran until the end of the academic year in July 2026.

My sample was of 103 schools spread throughout the Central Region. [Figure 2](#) shows their spatial distribution. Of these, 40 are low-fee private schools, and the other 63 are

government schools, all of which are part of the Cape Coast Metropolitan school district. For private schools to be eligible, they already had to be receiving TFLI, which required them to charge less than 400 Ghanaian Cedis per term (about 40 USD). All government schools were eligible, but I excluded schools designated for special needs students such as the school for the deaf before randomly assigning treatment statuses.

I randomly divided the 103 schools into four equally-sized groups: parent program and principal training, parent program only, principal training only, and control. [Table 1](#) summarizes the experimental design. I stratified the randomization by school type (private vs. government) and by beginning-of-year baseline reading ability measured by the letter sounds EGRA subtest. 16 private schools did not run baseline assessments. For these schools, I used the average reading ability of the previous year’s first-grade cohort.¹⁰ I targeted four schools per stratification cell. School sizes were highly varied at baseline, which adds a lot of noise to estimates of baseline reading ability. To account for this, I aggregated reading scores using empirical Bayes shrinkage ([Morris, 1982](#)). Empirical Bayes shrinkage calculates the overall mean reading ability for students in all schools, then allows deviations from that “grand mean” depending on how many students each school has. Schools with few students provide less information, so their derived mean can deviate less from the grand mean than larger schools can.

The randomization produced a balanced study across all four arms. Randomization inference F-tests of overall balance following [Kerwin, Rostom, and Sterck \(2024\)](#) produce non-statistically significant p -values ([Table 2](#)).

The 103 schools in my initial sample had 2,089 students. Of these, 1,680 (80%) were assessed at baseline while the rest were listed on class rosters but not assessed. Five schools (all private) closed before treatment began, and 8 private schools did not run end-of-year assessments—so I had 91 schools for endline data collection. [Table A1](#) shows the number of schools in each treatment arm post attrition. At these schools, 1,698 students were assessed at endline, which is a 19% attrition rate. Attrition rates were not differential across any study arms, and there was no difference in baseline covariates for children lost at endline

¹⁰ Three government schools did not run baseline tests, and I did not have data from a previous year, so they formed their own strata.

Table A2. I further test balance across the post-attrition sample and find the sample is balanced across students, teachers, principals, schools, and parents (Table A3, Table A4, Table A5, Table A6, and Table A7).

Table 1
Experimental Design

		Parent Program	
		<i>No</i>	<i>Yes</i>
Principal Training	<i>No</i>	26 schools	25 schools
	<i>Yes</i>	26 schools	26 schools

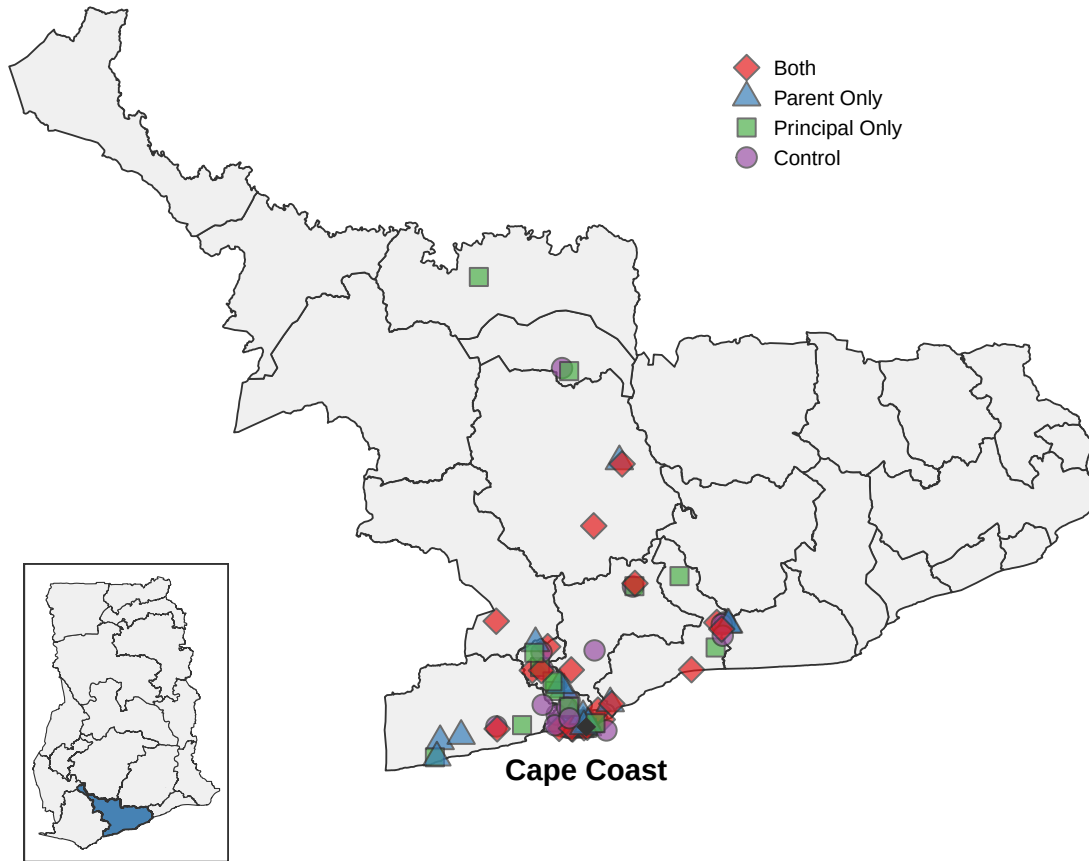
Notes: The 103 schools were randomly assigned to one of the four main study arms as shown in the table. See Figure 1 for the timeline and Table A1 for the post-attrition allocation of schools.

2.2 Data

I use data from three sources for this analysis: reading assessments, surveys, and administrative app use data. My primary data source is teacher-led assessments of reading fluency. As part of TFLI, teachers run regular reading assessments at the end of each school term and the beginning of the year, assessing various reading skills using EGRA subtests. In total, teachers ran four waves of data collection: two pre- and two post-treatment. The six standard subtests I use are letter names, letter sounds, decodable word reading, sight word reading, oral reading fluency, and reading comprehension. These subtests are arranged in order of increasing complexity. Letter sound knowledge, for example, has little value on its own but is a necessary component skill for building the economically valuable skills of reading fluency and comprehension.

My pre-analysis plan specified that letter sound knowledge would be my primary outcome; however, each assessment wave did not use the same subtests. The first and third waves used all six subtests, while the second and fourth only included reading fluency and comprehension subtests. I intended for the endline assessment wave to include all six subtests, but due to an issue with different versions of the app, only 30% of assessments included the remaining subtests, which is too small a sample to be meaningful. Appendix B analyzes the full set

Figure 2
Study Sample Schools



of outcomes for the students who received them. Letter sound knowledge is the most basic reading skill in the EGRA subtests I included, and so could have been an early indicator of the impacts of both programs. Since I do not have the letter sounds data at endline, I use oral reading fluency as my primary outcome since it is the most basic of the subtests actually run.

Teachers conducted the EGRA assessments orally and individually with each student using SmartCoach. Teacher-led assessments have a high correlation with those run by trained enumerators. The correlation is especially strong for ranking students' reading ability within a class, and there is no evidence that teachers strategically skip weaker students ([Andersen](#)

et al., 2026a). Teacher-led assessments can be a cost-effective alternative to collecting data with trained enumerators, but not all teachers are equally responsive to running all assessments. Most government schools ran all four waves of assessment, but private schools had much lower completion rates for the second and third assessment waves during the middle of the year and required more reminders from ITG staff.

I supplement this data with surveys of students, parents, teachers, and principals. Teachers ran a six-question student survey directly following the final EGRA assessment of the year. It asked students about their study practices at home with a caregiver. The survey was necessarily brief so teachers had time to assess the full class.

Trained enumerators conducted parent surveys over the phone. This survey asked parents about their at-home learning practices, including time spent reading with or to their children, whether they spent money on school-related expenses including books, and their beliefs on the importance of reading and the value of phonics-based instruction for achieving those goals.

Parent surveys ran from April to June 2026, but faced many difficulties. Many parents believed that the phone calls were scams or, in extreme cases, that the enumerators were collecting demographic and socioeconomic information so they could rob the parents' house. Because of this, only 47% of students in the endline sample had a parent survey. However, I do not find evidence for differential attrition for students without a parent survey (Table A8), and parents who did receive it have balanced covariates across all four study arms (Table A7).

The teacher and principal surveys measured demographics, professional experience, as well as their interactions with each other and parents. I also asked about their confidence in teaching or coaching and their work environment. The survey was administered to first-grade teachers in all schools, but only government school principals received a survey.

Finally, I collect administrative data. This data primarily measures management practices such as the number of principal coaching sessions and their content. The data also includes attendance at parent meetings.

2.3 Program Implementation & Uptake

The first wave of EGRA assessments took place from September to October 2025. Students could read 5.9 words per minute and answered 0.2 reading comprehension questions correctly out of three, with minimal differences between arms. There was significant growth in both measures at the second baseline EGRA assessment wave in December at the end of the first school term. Overall, students could read 18.5 words per minute and answer 0.48 reading comprehension questions correctly, which remained balanced across study arms.¹¹ Schools had an average of 23 enrolled first-grade students at the beginning of the year, of which 48% were male, and 52% were female. This is summarized in [Table 2](#)

I measure take-up for the parent intervention using attendance at each meeting. I also ask on the survey if they received the literacy workbook. 47 of the 51 schools assigned to receive the parent intervention attended the training. Of these, 36 collected attendance sheets at the first parent meeting while 30 collected attendance at the second. Some of the schools missing an attendance sheet did not run the meeting, but anecdotal reports from ITG suggest that many did run the meeting but failed to take attendance. Overall, 49% of eligible students had a parent attend meeting 1, 30% attended meeting 2, and 60% attended at least one meeting. Beyond the schools that held meetings but did not return an attendance sheet, several schools returned sheets that were illegible. I could not match the parents to their children, so these are likely a lower bound of the true attendance. Of the schools with attendance data, 49% of children had a parent attend the first meeting while 42% had a parent attend the second meeting. There is a significant lack of overlap between these groups: only 26% of children had a parent attend both meetings. Overall, 60% of families attended at least one. Due to the relatively low level of attendance, I report treatment-on-the-treated effects using attendance and workbook possession as an instrument. Implementation statistics for both interventions are summarized in [Table 3](#).

Children whose parents would attend a meeting were markedly different than those whose parents would not attend any meetings. The differences are summarized in [Table A9](#). They

¹¹ 639 fewer students took the second EGRA assessment, mostly due to some private schools not running any. However, within the panel of students who received an assessment in both waves, the levels of reading ability and growth between baseline assessment waves was identical to the overall average.

were more likely to be female, and far less likely to have taken the beginning-of-the-year reading assessment. However, their parents did not have many differences. They had no detectable differences in levels of schooling, assets owned, or number of children. They were slightly younger and slightly more likely to live with their child. The main difference is that attending parents were more engaged: they were 16.5 percentage points more likely to have completed a survey later in the year. At the first baseline, this difference did not manifest in reading ability, but by the end of the first term (still pre-treatment), children whose parents would eventually attend could read 2.9 more words per minute and answered 0.16 more reading comprehension questions correctly, as shown in [Figure 5](#).¹²

I measure implementation for the principal intervention using attendance at the training and the score on a brief post-training knowledge quiz. 86.5% of schools randomized to receive the principal intervention attended the first training. Three of these closed before training occurred, so of the open schools, 91.8% were represented. 78.8% of schools attended the second training in April. These groups mostly overlapped, so 88.5% attended at least one meeting. Schools generally sent multiple representatives: on average, 2.2 people attended per school. These were mostly headteachers, along with assistant headteachers, and some schools sent their curriculum leaders. At the end of the training, the attendees took a brief knowledge quiz and scored 94.9% correctly on average. Overall, implementation for the principal intervention was strong. Before the training, principals ran 2.1 lesson observations in their first-grade classes before the intervention began, and this was balanced across study arms.

3 Empirical Strategy

My main estimating equation is:

$$Y_{ij} = \beta_0 + \beta_1 \text{Parent}_j + \beta_2 \text{Principal}_j + \beta_3 \text{Both}_j + Z_j' \tau + X_i' \gamma + \epsilon_{ij} \quad (1)$$

¹² These estimates are regression adjusted for age-by-gender fixed effects and baseline reading ability.

Table 2
Balance Table

	(1) Control Mean (S.E.)	(2) Parent Diff. (<i>p</i> -value)	(3) Principal Diff. (<i>p</i> -value)	(4) Parent and Principal Diff. (<i>p</i> -value)	(5) Obs.
Panel A: Student-Level Variables					
Male	0.509 (0.040)	-0.038 (0.429)	0.046 (0.499)	-0.042 (0.672)	2,089
Student Age (Years)	6.615 (0.072)	0.029 (0.837)	0.253** (0.046)	-0.127 (0.435)	935
Missing Baseline Assessment	0.132 (0.039)	0.033 (0.331)	-0.012 (0.665)	0.119** (0.037)	2,089
Baseline Reading Index	-0.000 (0.087)	-0.036 (0.716)	0.041 (0.706)	-0.000 (0.999)	1,680
Missing Second Baseline Assessment	0.495 (0.096)	-0.047 (0.628)	0.004 (0.963)	-0.108 (0.296)	2,089
Second Baseline ORF	18.485 (1.020)	0.527 (0.865)	0.869 (0.833)	-2.454 (0.448)	1,041
Second Baseline Reading Comp.	0.478 (0.026)	0.099 (0.467)	0.274 (0.153)	-0.106 (0.293)	1,041
Joint F-stat R.I. <i>p</i> -value		1.32 [0.797]	7.61 [0.068]	4.31 [0.203]	
Panel B: School-Level Variables					
First-Grade Students on Rolls	23.360 (3.396)	-1.797 (0.719)	-6.912 (0.181)	-1.229 (0.861)	99
Share Male	0.477 (0.041)	0.014 (0.845)	0.055 (0.409)	0.019 (0.836)	99
First-Grade Lesson Observations Before Training	2.080 (0.551)	0.122 (0.851)	0.735 (0.325)	0.470 (0.420)	98
Share Missing Baseline Assessment	0.232 (0.071)	0.076 (0.148)	0.016 (0.620)	0.076 (0.121)	99
School Mean Baseline Reading Index	-0.002 (0.119)	0.035 (0.761)	0.017 (0.900)	-0.021 (0.896)	84
Share Missing Second Baseline Assessment	0.536 (0.087)	0.037 (0.768)	0.058 (0.604)	0.037 (0.796)	99
School Mean Second Baseline ORF	30.245 (13.094)	3.839 (0.512)	0.416 (0.959)	-2.343 (0.438)	55
School Mean Second Baseline Reading Comp.	0.587 (0.178)	0.246 (0.239)	0.344 (0.259)	-0.094 (0.367)	55
Joint F-stat R.I. <i>p</i> -value		2.16 [0.710]	2.01 [0.734]	7.02 [0.260]	

Notes: Sample is all students who were enrolled in one of the 103 study schools at the beginning of the 2025-26 academic year. Baseline refers to EGRA exams run in September 2026. Second baseline refers to EGRA exams run in November-December 2025 before treatment. The reading index is constructed by taking the first principal component of the control group for all EGRA subtests: letter sounds, decodable words, sight words, ORF, and reading comp. and applying those weights to each treatment arm as well. ORF stands for oral reading fluency. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses (). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table 3
Program Take-up

	(1) Share / Mean	(2) Obs.
Panel A: Parent Program		
School Staff Trained to Deliver Meetings	0.979	48
Attendance Sheet Returned, Meeting 1	0.771	48
Attendance Sheet Returned, Meeting 2	0.625	48
Parent Attended Meeting 1	0.489	802
Parent Attended Meeting 2	0.308	802
Parent Attended At Least One Meeting	0.603	802
Parent Attended Both Meetings	0.193	802
Panel B: Principal Program		
School Attended Training, Session 1	0.865	52
School Attended Training, Session 2	0.788	52
School Attended At Least One Session	0.885	52
Average Attendees per Attending School	2.23	86
Knowledge Quiz Score [0-10]	9.49	86

Notes: All entries are unadjusted shares or means. Panel A covers the parent program. The first three rows are shares of the 48 parent-program schools (parent-only and combined arms) appearing in the endline data. The attendance rows are student-level shares over a single, conservative denominator: the 802 students at schools that returned a Meeting 1 sign-in sheet. Panel B covers the principal program. The training rows are shares of all 52 randomized principal-program schools (principal-only and combined arms) with at least one representative at each training session, held in January and April 2026; schools that do not appear in the endline data are counted as not attending. Average attendees per attending school is calculated by dividing the 192 attendees recorded in the surveys administered at the sessions by the 86 school-session attendances (45 schools at the first session and 41 at the second). The knowledge quiz was administered during the training sessions; the final row reports the attendee-weighted mean score among the 86 attendees who completed it. Observation counts in column (2) accordingly refer to schools in the first three rows of each panel, students in the attendance rows, school-sessions in the attendees row, and quiz takers in the final row.

In this equation, Y_{ij} is the outcome of interest and i indexes students, who are nested within their original schools indexed by j . $Parent_j$ is an indicator for a school receiving the parental engagement intervention, $Principal_j$ is an indicator for a school receiving the principal training, and $Both_j$ is an indicator for a school receiving both interventions. Z_j is a vector of indicators for the stratification cells used in the lottery that assigned the schools to treatment arms. X_i is a vector of control variables. I include the following variables as controls: first and second baseline values of Y_{ij} if available, an indicator for a student being male, indicators for each value of age in years, the interactions between the two, and indicators for parent education level. I winsorize student age at the 5th and 95th percentiles. I set missing baseline scores to zero and pair them with an indicator for having no assessment in that wave; missing values of the categorical controls are coded as separate categories and included when constructing the indicators and interactions so that these missings are dummied out in a fully parametric way. Oral reading fluency and reading comprehension both have two baselines; I control for both baselines separately. β_1 and β_2 capture the average impact of the two individual interventions when they are implemented alone relative to the control group, while β_3 captures the average impact of receiving both. I also report α_3 from [Equation 2](#), which is the additional effect of combining the two programs above additivity.

$$Y_{ij} = \alpha_0 + \alpha_1 Parent_j + \alpha_2 Principal_j + \alpha_3 Parent_j \cdot Principal_j + Z_j' \tau + X_i' \gamma + \epsilon_{ij} \quad (2)$$

In stratified experiments with small strata, standard errors should be clustered at the level of stratification rather than the level of randomization to optimize power and the size of hypothesis tests ([de Chaisemartin and Ramirez-Cuellar, 2024](#)). Accordingly, I report standard errors clustered this way since I have an average of four schools per stratification cell.

I have two families of outcomes: literacy and parental engagement. My primary outcome for literacy is oral reading fluency. Because this is my only primary outcome, I do not correct for multiple hypothesis tests. I also report treatment effects on reading comprehension for all students and for each of the other EGRA subtests in [Appendix B](#) for the sample of students that received them.

My second family of outcomes is parental engagement. My primary outcome for parental engagement is how many days parents report reading with their student in a typical week. While there are many facets to parental engagement, this is most closely linked to the treatment. I also report effects on other reported parenting practices such as the number of days in a typical week a parent tells their child a story, practices reading, math, or writing with their child, talks about what the child is learning in school, works on any learning activities at home, checks their child’s homework, and meets with their child’s teacher or principal. Additionally, I report how much they spend on school-related expenses (excluding school fees), and whether they purchase any books during a typical school term.

I also use [Equation 1](#) to estimate effects from the student, parent, teacher, and school leader surveys, and all administrative data.

In [Table C1](#), I show additional estimates without baseline controls, without stratification cells, and adjusting stratification cell assignments to correct for full schools attriting from the study.¹³ Although I do not find evidence of differential attrition ([Table A2](#)), I report [Lee \(2005\)](#) bounds in [Table C2](#). My pre-analysis plan specifies that I would use one-sided hypotheses for the estimates of the treatment effects for each individual program. Out of an abundance of caution and because the point estimate for the principal intervention is negative, I use standard two-sided hypothesis tests throughout this paper. [Table C3](#) presents estimates for my main results with one-sided tests.

Given the relatively limited take-up of the parent intervention, β_i should be interpreted as measuring the impact of having the intervention implemented in children’s schools. Because of this, I also estimate the treatment-on-the-treated effect using the following two stage least squares regression using $Parent_j$ as an instrument for $Compliance_{ij}$:

$$Y_{ij} = \beta_0 + \beta_1 Compliance_{ij} + \beta_2 Principal_j + \beta_3 Both_j + Z_j' \tau + X_i' \gamma + \epsilon_{ij} \quad (3)$$

I construct three versions of *Compliance*. First, I construct an index of the following three indicators: (i) attendance at the first parent meeting, (ii) attendance at the second parent

¹³ When full schools attrit, this can lead to there being no variation in treatment status within a stratification cell. I reassign schools from that stratification cell to a stratification cell with a value one lower than the one assigned. E.G., schools in stratification cell 5 would be assigned to cell 4.

meeting, (iii) having the literacy workbook at home. Second, because of low response rates to the parent survey, I construct an index of just (i) and (ii). Finally, I use an indicator of attendance at either of the meetings. Together, these instruments allow me to estimate how well the parent program would have worked with full compliance under the assumption that the effects of the intervention work only through compliance with the program.

4 Results

Table 4 presents the impact of the programs on the children’s literacy outcomes, which are my primary outcomes. The parent intervention has large, positive effects on both reading fluency and comprehension. Reading fluency increases by 0.33 standard deviations ($p = 0.045$) while comprehension increases by an imprecisely estimated 0.16 SDs ($p = 0.215$). Respectively, these are equivalent to 0.39 and 0.2 additional years of schooling for the control group.¹⁴ Control-group students are not receiving status-quo instruction for Ghana. They are instead receiving an effective structured pedagogy program (Andersen et al., 2026b). Control-group fluency growth is much faster than typical students in this setting: control-group students grow 0.85 SDs in fluency in a year versus 0.22 SDs under Ghanaian status-quo literacy instruction (Evans and Yuan, 2019).¹⁵ Under status-quo growth rates, the parent program increased reading fluency and comprehension by an additional 1.5 and 0.7 years, respectively.

While the parent intervention’s main effects are on reading fluency, the principal intervention had its largest effects on reading comprehension. Reading comprehension decreases by 0.26 SDs ($p = 0.048$). The effect on reading fluency is smaller and imprecisely estimated. These estimates should also be interpreted as relative to better-than-status-quo instruction.

The fourth row of Table 4 shows my estimates of the interaction between the two interventions.¹⁶ Because the principal intervention negatively affected reading outcomes, the

¹⁴ Because of stop rules used during the baseline EGRA, only 49% of students reached the subtests for oral reading fluency and 34% reached reading comprehension. For this calculation, I impute those students as scoring 0 if they took the baseline EGRA but stopped before reaching a particular subtest. Using only students who completed the full EGRA gives 0.3 and 0.14 additional years of schooling estimates for reading fluency and comprehension respectively. Restricting to complete EGRA cases selects for only the strongest readers at baseline, so these are likely an overestimate of control-group growth and thus an underestimate of the number of additional years of schooling.

¹⁵ Control-group growth is 0.79 SDs in reading comprehension.

¹⁶ Mechanically, the estimated effect of receiving both programs (row 3) equals the sum of the individual-

typical interpretation of this term flips: if home and school inputs are complements, a program that reduces school quality also lowers the returns to parental investment, and the combination would amplify the reduction in reading scores. A positive interaction instead implies that one program offsets the other either because the same home investment is more productive where instruction quality has fallen, or because households invest more in response to the damage. The point estimate for reading comprehension is positive, implying the parent program offset the reduction in reading comprehension in principal-intervention schools, or equivalently, that the parent program was more effective in schools that also received the principal intervention; [Table D1](#) reports these conditional effects. The interaction is not statistically different from zero for reading comprehension, but the 95% confidence interval rules out interactions more negative than -0.11 SDs. I find no evidence that parental disengagement amplified the principal program’s reduction in reading scores, and any such effect would be small in magnitude. For reading fluency, the interaction is an imprecisely estimated null (-0.03 SDs).

Although I find no differential attrition across the study arms ([Table A2](#)), I present estimates of Lee bounds ([Lee, 2005](#)) in [Table C2](#). The parent intervention has statistically significant effects on reading fluency, and the upper bound stays significant while the lower bound loses significance. This is mechanically true. The p -value of my primary estimates is very close to the 5% significance threshold, so any lower bound would drop significance. Similarly, the lower bound (because the effect is negative) for the principal intervention’s effect on reading comprehension remains significant while the upper bound loses significance.

4.1 Heterogeneity

Positive average effects from literacy programs can mask some students falling behind ([Buhl-Wiggers et al., 2024](#)). The parent intervention does not exhibit this concern, and its gains are broad-based. I find no distinguishable difference in its treatment effect by gender, school type, teacher experience, or parental education.¹⁷ Its gains appear to be equally spread across the distribution of reading ability. In contrast, the principal intervention shows negative

program estimates (rows 1 and 2) plus the interaction (row 4).

¹⁷ Gender is the only subgroup I pre-specified, so the rest should be interpreted as exploratory.

Table 4
Program Impact on Test Scores

	(1) Oral Reading Fluency	(2)	(3) Reading Comprehension	(4)
	CWPM	SDs	Score [0-3]	SDs
Parent	9.224	0.331	0.176	0.159
S.E.	(4.746)	(0.170)	(0.133)	(0.121)
R.I. <i>p</i> -value	[0.045]**		[0.215]	
Principal	-2.959	-0.106	-0.290	-0.262
S.E.	(3.037)	(0.109)	(0.111)	(0.101)
R.I. <i>p</i> -value	[0.528]		[0.048]**	
Parent and Principal	5.383	0.193	0.102	0.093
S.E.	(4.028)	(0.144)	(0.118)	(0.107)
R.I. <i>p</i> -value	[0.219]		[0.468]	
Interaction	-0.882	-0.032	0.217	0.196
95% CI	[-12.12, 10.35]	[-0.43, 0.37]	[-0.12, 0.56]	[-0.11, 0.50]
R.I. <i>p</i> -value	[0.887]		[0.316]	
N (# students)	1,698		1,698	
C (# strat. cells)	26		26	
Adjusted R ²	0.300		0.334	
Control-group values				
Mean	30.320		1.127	
SD	27.874		1.106	

Notes: Sample is all students with a valid endline literacy assessment, enrolled in one of the 103 study schools at the beginning of the 2025/26 academic year. All schools received the Inspiring Teachers structured pedagogy program, so estimates identify effects relative to strong classroom instruction. Columns (1) and (3) report effects in the outcome’s natural units; columns (2) and (4) in standard deviations of the control-group distribution. The first three rows report effects of each treatment arm relative to the control group. The Interaction row reports the difference between the combined-program effect and the sum of the two individual effects. [Table D1](#) reports each program’s effect conditional on the other program. All regressions control for the baseline and second-baseline values of the dependent variable, with missing values zeroed out and indicated, a full set of age-category-by-gender interactions, parent education indicators, and stratification cell fixed effects. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses; the Interaction standard error is computed from the full covariance matrix of the arm coefficients. Randomization-inference p -values in brackets permute the four-arm assignment within the original stratification cells, clustered by school (1,000 draws); all statistics use the same permutation draws. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

treatment effects concentrated among girls and weaker readers.

In [Table 5](#), I show [Equation 1](#) augmented with indicators for various dimensions of heterogeneity. The parent intervention has no statistically significant interactions. Because these are demanding tests, the standard errors on the interactions are generally large. I cannot rule out economically meaningful heterogeneous treatment effects, but my results suggest that disadvantaged groups still experience large gains in reading. Students whose

parents completed less education than junior high school gain 0.67 SDs in fluency ($p = 0.05$) and 0.43 SDs in comprehension ($p = 0.07$). Similarly, boys gain 0.33 and 0.23 SDs in fluency and comprehension respectively, despite control-group boys reading 8.9 words per minute and answering 0.45 fewer comprehension questions correctly than girls.

The principal intervention does show differentially harmful effects. Its negative effect on reading comprehension is larger for girls by 0.29 SDs ($p = 0.04$). Similarly, although the difference is imprecisely estimated, government schools have a larger magnitude loss in reading comprehension. Comprehension decreases by 0.33 SDs in government schools, while in private schools there was no effect, although this difference is imprecisely estimated ($p = 0.32$).

The parent intervention’s effects appear to be consistent throughout students’ ability distribution. [Figure 3](#) and [Figure 4](#) show effects from quantile ([Koenker and Xiao, 2003](#)) and distribution ([Chernozhukov, Fernández-Val, and Melly, 2013](#)) regressions on reading fluency and comprehension respectively.¹⁸ I do not show quantile treatment effects (QTEs) for reading comprehension because the subtest has only three points in its support. Both reading fluency and comprehension show large gains throughout their distributions. The gains for reading fluency grow at larger quantiles but are positive everywhere. The distribution regressions show the probability of students crossing reading benchmarks. Students are more likely to cross every benchmark for both fluency and comprehension. In particular, students are 14 percentage points more likely to be able to read at least 45 words per minute ($p = 0.05$), which is a commonly used benchmark for full fluency ([Abadzi, 2012](#)). The proportions are larger and statistically significant for students reading between 30 and 60 words per minute.

The principal intervention has more negative fluency QTEs for moderate quantiles, which move toward no effect at the top and bottom of the distribution. The distribution regressions show a similar shape. The proportion of students able to read at least 20 words per minute is 12 percentage points lower with the principal intervention compared to the control group. The area of the distribution that sees the biggest reduction is where the majority of students sit. Similarly, the proportion of students able to answer two or three reading comprehension questions correctly decreases by 14 percentage points. Overall, the principal intervention

¹⁸ See [Andersen et al. \(2026b\)](#) for an intuitive explanation of the difference between the two methods.

appears to harm moderate- to high-reading-ability students more than weaker ones. This is consistent with its average effects being concentrated in reading comprehension, which is a more advanced skill than reading fluency.

I show treatment effects decomposed by students' initial test scores in [Appendix E](#), and they show similar patterns. However, because of gaps in baseline coverage for reading fluency and comprehension, the distribution and quantile effects are my preferred measure of distributional heterogeneity.

Figure 3
Distributional Effects on Oral Reading Fluency

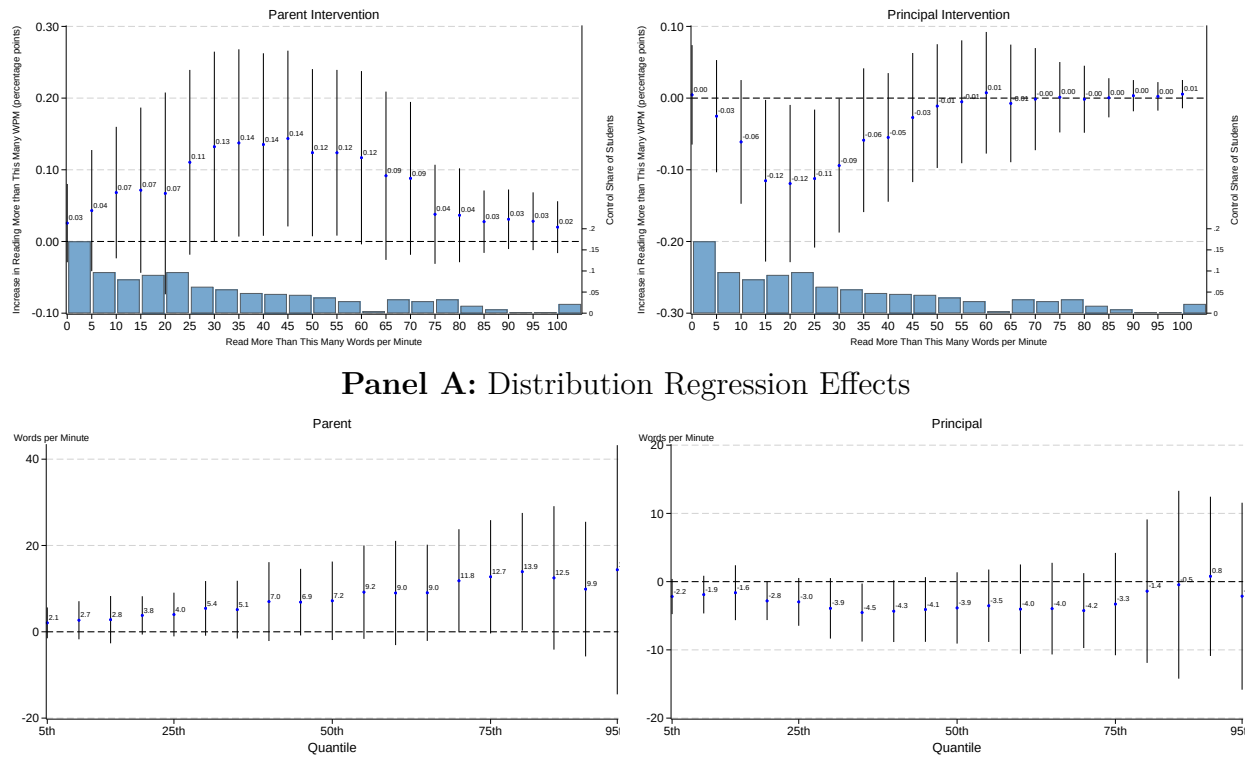


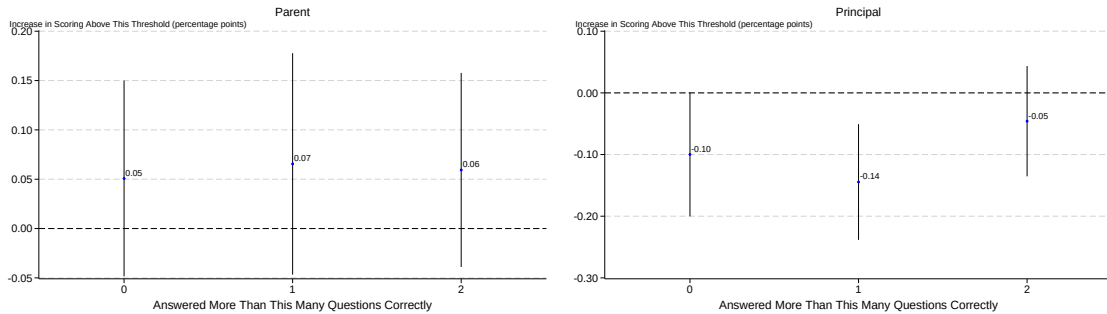
Table 5
Heterogeneity Table

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Interaction with...</i>	Girl		Private School		Teacher Experience		Parent Completed JHS+	
<i>Dependent variable:</i>	ORF (SDs)	Reading Comp. (SDs)	ORF (SDs)	Reading Comp. (SDs)	ORF (SDs)	Reading Comp. (SDs)	ORF (SDs)	Reading Comp. (SDs)
Parent $\times$...	0.001	-0.128	0.275	0.388	-0.046	-0.037	-0.310	-0.188
S.E.	(0.144)	(0.151)	(0.330)	(0.280)	(0.019)	(0.015)	(0.351)	(0.233)
R.I. <i>p</i> -value	[0.995]	[0.342]	[0.525]	[0.242]	[0.174]	[0.175]	[0.300]	[0.427]
Principal $\times$...	-0.104	-0.293	0.136	0.396	-0.018	-0.013	0.023	-0.066
S.E.	(0.143)	(0.168)	(0.194)	(0.205)	(0.015)	(0.012)	(0.207)	(0.208)
R.I. <i>p</i> -value	[0.474]	[0.036]**	[0.783]	[0.315]	[0.544]	[0.601]	[0.946]	[0.774]
Parent and Principal $\times$...	0.071	-0.160	0.398	0.446	-0.085	-0.077	-0.043	-0.016
S.E.	(0.134)	(0.106)	(0.333)	(0.259)	(0.023)	(0.017)	(0.202)	(0.185)
R.I. <i>p</i> -value	[0.637]	[0.261]	[0.385]	[0.206]	[0.002]***	[0.002]***	[0.880]	[0.944]
Parent	0.328	0.226	0.286	0.093	0.426	0.221	0.670	0.426
S.E.	(0.195)	(0.151)	(0.195)	(0.117)	(0.167)	(0.117)	(0.423)	(0.243)
R.I. <i>p</i> -value	[0.060]*	[0.101]	[0.125]	[0.501]	[0.069]*	[0.225]	[0.051]*	[0.066]*
Principal	-0.057	-0.124	-0.114	-0.328	-0.044	-0.217	-0.097	-0.064
S.E.	(0.132)	(0.132)	(0.134)	(0.115)	(0.151)	(0.119)	(0.239)	(0.196)
R.I. <i>p</i> -value	[0.726]	[0.374]	[0.532]	[0.015]**	[0.859]	[0.209]	[0.816]	[0.780]
Parent and Principal	0.161	0.190	0.122	0.021	0.197	0.062	0.202	0.045
S.E.	(0.164)	(0.106)	(0.160)	(0.121)	(0.156)	(0.117)	(0.212)	(0.167)
R.I. <i>p</i> -value	[0.354]	[0.158]	[0.521]	[0.880]	[0.397]	[0.667]	[0.611]	[0.847]
N (# students)	1,698	1,698	1,698	1,698	1,380	1,380	710	710
C (# strat. cells)	26	26	26	26	24	24	26	26
Adjusted R ²	0.300	0.334	0.302	0.337	0.331	0.383	0.349	0.357

Notes: Sample is all students with a valid endline literacy assessment, enrolled in one of the 103 study schools at the beginning of the 2025/26 academic year; columns (5)–(8) are further restricted by moderator availability, to students at the 69 schools with an endline teacher survey (columns 5 and 6) and to students whose surveyed parent reported an education level (columns 7 and 8). All schools received the Inspiring Teachers structured pedagogy program, so estimates identify effects relative to strong classroom instruction. All outcomes are in standard deviations of the control-group distribution. Each column pair reports a single regression of the outcome on the three treatment-arm indicators, the column’s moderator, and their interactions. The first three rows report how each arm’s effect differs with the moderator; the next three report each arm’s effect for the moderator-zero group. The effect for the moderator-one group is the sum of the main effect and the interaction. All regressions control for the baseline and second-baseline values of the dependent variable, with missing values zeroed out and indicated, a full set of age-category-by-gender interactions (age categories only in columns 1 and 2, where the moderator replaces the gender control), parent education indicators (omitted in columns 7 and 8, where the moderator is a coarsening of that control), and stratification cell fixed effects. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses. Randomization-inference *p*-values in brackets permute the four-arm assignment within the original stratification cells, clustered by school (1,000 draws); all statistics use the same permutation draws. Gender is the heterogeneity dimension pre-specified in the analysis plan; the remaining moderators are exploratory. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Note: Notes: Sample is all students with a valid endline literacy assessment, enrolled in one of the 103 study schools at the beginning of the 2025/26 academic year. Each panel reports estimates for one treatment arm relative to the control group, from the same four-arm specification as the main results. Panel A reports distribution regressions: at each 5-word threshold from 0 to 100, the outcome is an indicator for reading more than that many words per minute, and the plotted point is the arm’s effect on the probability of clearing the threshold, in percentage points. Bars show the share of control-group students at each point of the fluency distribution (right axis). Panel B reports quantile treatment effects: each point is the arm’s effect on the indicated quantile of the endline fluency distribution, in words per minute, estimated by quantile regression at the 5th through 95th percentiles. Quantiles refer to the distribution of the outcome, not to students’ positions in the baseline distribution. All regressions control for the baseline and second-baseline values of the dependent variable, with missing values zeroed out and indicated, a full set of age-category-by-gender interactions, parent education indicators, and stratification cell fixed effects. Whiskers are 95% confidence intervals based on standard errors clustered by stratification cell. A bootstrap test cannot reject that either arm’s quantile profile is a uniform shift of the distribution (parent: $p = 0.14$; principal: $p = 0.92$; see text).

Figure 4
Distributional Effects on Reading Comprehension



Panel A: Distribution Regression Effects

Note: Sample is all students with a valid endline literacy assessment, enrolled in one of the 103 study schools at the beginning of the 2025/26 academic year. Each panel reports estimates for one treatment arm relative to the control group, from the same four-arm specification as the main results. At each threshold, the outcome is an indicator for answering more than that many of the three reading comprehension questions correctly, and the plotted point is the arm’s effect on the probability of clearing the threshold, in percentage points. No quantile panel accompanies this figure because comprehension is a four-point score, for which quantile treatment effects are not well defined. All regressions control for the baseline and second-baseline values of the dependent variable with missing values zeroed out and indicated, a full set of age-category-by-gender interactions, parent education indicators, and stratification cell fixed effects. Whiskers are 95% confidence intervals based on standard errors clustered by stratification cell.

4.2 Mechanisms

4.2.1 Parent Program

There are two main channels through which the two interventions could function: changing engagement at home or changing practices at school. The parent intervention works primarily through the first channel. [Table 6](#) shows parents’ reports of their level of engagement with their children’s schooling. Overall, my index of parental engagement is not affected by the parent intervention. If anything, parent-reported engagement at home decreases. Although not statistically significant, the parent intervention reduces the number of days parents read to their children in a typical week by 21%. The workbook also functions as a substitute for other book purchases for their students: parents decrease their book purchases by 11 percentage points. Despite no effects on engagement at home, teachers report parents become *more* engaged at school ([Table 9](#)). Teachers in the parent arm report a 47% increase in the number of one-on-one parent meetings they held. The survey also asked them how much they agreed that parents have the needed knowledge to support reading at home, and the parent intervention arm raised agreement by 0.65 points.

Supporting children’s learning at home is not limited to parents. To account for this, [Table 7](#) reports students’ responses to survey questions about their home engagement environment. Students experience an increase in family engagement. Overall, engagement increases by 0.56 SDs ($p = 0.09$). Family members increase the number of days reading with students by 33% ($p = 0.16$), the number of days explicitly practicing reading by 35% ($p = 0.21$), and the number of days talking about school by 45% ($p = 0.06$). Crucially, the parent intervention causes a 25% shift away from siblings as a primary supervisor of learning activities.¹⁹ Often in developing countries, this job falls to siblings ([Jakiela et al., 2025](#)). In my setting, the plurality of students in the control group (48%) report that a sibling is their primary supervisor for learning activities at home ([Table D2](#)). Despite the intervention being targeted at parents, its effects may not show up in parents’ actions. Sending the literacy workbook home is a plausible channel through which siblings could improve their supervision of school activities despite not attending the meetings. Interventions that survey

¹⁹ The prevalence of sibling supervisors remains quite high at 33% of children after this shift.

only parents to find changes in engagement may be missing the whole picture.

As I discuss in [Section 2.3](#), attendance at the parent meetings was relatively low. To account for this, I measure implementation of the parent intervention’s crucial components: meeting attendance and workbook delivery. [Table 8](#) reports these results using three measures of take-up. Columns 1 and 2 use an index of attending both meetings and receiving a workbook. This is my pre-specified implementation index, but it is not my preferred estimate because of low reporting rates for workbook delivery. Columns 3 and 4 use attendance at each meeting separately as their implementation measure, and columns 5 and 6 use my preferred measure of attendance at either meeting. Using my preferred measure of implementation, Panel B shows that there is a high correlation between meeting attendance and both reading fluency and comprehension. Panel C shows estimates of the effect of full meeting attendance using assignment to the parent intervention study arm as an instrument for program compliance. This estimates the increase in reading ability under full meeting attendance under the assumption that the parent intervention only affects reading ability through the meetings. My estimates suggest that full attendance would increase fluency by 0.58 SDs and reading comprehension by 0.28 SDs, although the latter is not statistically significant. If realized, the fluency effect size would sit about the 90th percentile of literacy interventions ([Evans and Yuan, 2022](#)). Columns 1 and 2 show that using the workbook as an additional channel (in the smaller sample of parents who responded to that question on the survey) full compliance would increase fluency by 1.47 SDs and comprehension by 1 SD, both of which would sit above the 99th percentile of effect sizes.

The estimates in Panel C report treatment effects for compliers with the instrument. I am able to describe multiple dimensions of who the complier set is. [Table A9](#) summarizes demographic differences among meeting attenders versus non-attenders. Students of parents who attend at least one meeting are more often female, and less likely to have missed taking a baseline assessment (a rough proxy for attendance). Parents who attend a meeting are younger, more likely to live with a child, and more likely to respond to my parent survey. Not only do attenders differ demographically, but they also differ on baseline reading level and pre-treatment reading growth rate. At the beginning of the year, children of parents who would eventually attend at least one meeting could answer 0.09 more reading comprehension

questions correctly. The divergence is not just in the level of reading ability but also in the growth rate. By the end of term 1 (before treatment began), those students could read 2.9 more words per minute and answer 0.16 more reading comprehension questions correctly. However, once treatment started, the divergence between attenders and non-attenders stopped. Between midline and endline data collection, attenders retained the same 5.8 words per minute gap over non-attenders, and slightly decreased their lead in reading comprehension from 0.31 to 0.27 more correct answers. Since the gap stopped growing, it is likely that the intervention realized its gains through channels beyond the meetings, such as the workbooks reaching home or spillovers between children or their families. Thus, the two-stage least squares estimates represent an upper bound to the program’s gains with full implementation.

4.2.2 Principal Program

The principal intervention operates primarily through changes in teaching practices. The theory of change for the intervention was that the principal training would increase the frequency that principals observe their teachers’ lessons and give them feedback. Teachers could use this to improve their quality of instruction and adherence to the structured pedagogy program, which would eventually lead to improved reading outcomes for students. I show evidence that this broke down at multiple stages. [Table 9](#) summarizes my findings. First, lesson observations appeared to increase using administrative data, but this is not corroborated by self-reports from teachers or principals. The number of lesson observations logged on the SmartCoach app increases by 62% ($p = 0.1$), but the number of lesson observations reported by teachers or principals does not increase. In the control group, the number of lesson observations is relatively equal across the different reporting methods. I attribute this discrepancy to the intervention changing principals’ tendency to use the app rather than changing their behavior. Part of the intervention encouraged using SmartCoach as an administrative tool more often, and this reporting pattern is consistent with the training causing principals to substitute informal, non-tracked coaching sessions for sessions logged on the app. A phantom increase in coaching visits cannot however explain why the program decreases reading comprehension.

The second step in the theory of change is that increased coaching improves teaching quality. I show that teachers changed their in-classroom behavior in response to the intervention, but not in a beneficial way. Teachers were 26% less likely to report frequently grading their students' workbooks ($p = 0.01$), and 36% less likely to adapt their lessons ($p = 0.19$). While reducing adaptation of high-quality lesson materials can be beneficial, it can also reduce the amount of tailoring of instruction teachers do to match the level of their students. Teachers also report feeling stressed by feedback 98% more often than in the control group ($p = 0.08$) and were less likely to agree that their head teacher helped improve their instruction quality. This pattern of teacher behavior, despite no increase in the number of coaching sessions, is again consistent with principals substituting informal sessions for the structured, logged coaching sessions the training prescribed. When teaching quality is measured and tracked, principals may have felt pressured to increase it, which fell harmfully onto teachers and students. This harm had a larger effect on reading comprehension because it is a more complex skill to teach and requires more context from teachers.

I also show suggestive evidence that within the principal intervention arm, students and their families increased their home engagement to compensate for the decreased quality of instruction at schools. Students report a shift of supervision at home from siblings to parents of 36% (larger than the shift in the parent arm), and overall student-reported family engagement increases by 0.39 SDs ($p = 0.24$). This is driven by an increase in the number of days families read with students, tell students a story, practice reading together, and talk about school. Unlike the parent intervention, parents do not increase their number of meetings with teachers or principals.

4.2.3 Program Interaction

I find suggestive evidence that parents offset the decreased school quality by engaging more. In schools that receive both interventions, parents discuss reading fluency test results with principals (but not teachers) 44% more often than in the control group ($p = 0.01$). Neither arm alone shifts the amount of discussions. When parents are empowered to seek out information because of the parent meetings, and the instruction quality drops, they do so frequently. Engagement reported by parents and students increases by 0.29 ($p = 0.25$) and

0.44 SDs ($p = 0.22$) respectively.²⁰

This pattern is consistent with concave production of literacy. Concavity implies that when instruction quality falls, the marginal return from home investment rises, so parents invest more. This would happen whenever school quality fell, but it would be amplified by the parent intervention that empowers parents. My results fit this model. When instruction quality decreases in the principal-only arm, parent engagement increases by 0.39 SDs ($p = 0.24$) from student reports and 0.2 SDs ($p = 0.41$) from parent reports. Concavity of investment explains why I find suggestive evidence that the programs complement each other for reading comprehension. The increased parent engagement offsets the decreased classroom instruction quality. Reading fluency does not exhibit this behavior because the principal program did not have as large a negative effect on it.

²⁰ However, children’s reports of family engagement decrease compared to the effects of the two programs individually. Students report talking about school about two days a week in schools that get both programs compared to 2.4 and 2.3 days a week in schools with the parent and principal intervention respectively. That is one fewer day than if the effects of the two programs stacked linearly ($p = 0.08$). Overall, students report 0.5 SDs less interaction compared to if the programs combined linearly ($p = 0.29$).

Table 6
Program Impact on Parent Engagement

	(1) Read to Child Days [0-7]	(2) Told a Story Days [0-7]	(3) Practiced Reading Days [0-7]	(4) Practiced Math Days [0-7]	(5) Practiced Writing Days [0-7]	(6) Talked About School Days [0-7]	(7) Checked Homework Days [0-7]	(8) Minutes on Learning Minutes	(9) Met Teacher or Leader 0/1	(10) School Expenses GHS	(11) Bought Books 0/1	(12) Has Workbook 0/1	(13) Engagement Index SDs	(14) School Quality 0-10	(15) Aspires Tertiary 0/1
Parent	-0.188	0.145	0.097	-0.026	-0.137	-0.211	-0.169	-10.660	0.067	13.883	-0.099	0.188	-0.025	-0.044	0.032
S.E.	(0.345)	(0.217)	(0.253)	(0.295)	(0.241)	(0.426)	(0.275)	(7.808)	(0.051)	(188.148)	(0.049)	(0.114)	(0.297)	(0.264)	(0.043)
R.I. <i>p</i> -value	[0.474]	[0.485]	[0.672]	[0.923]	[0.506]	[0.556]	[0.577]	[0.358]	[0.313]	[0.952]	[0.042]**	[0.112]	[0.921]	[0.881]	[0.565]
Principal	0.191	0.093	0.292	0.299	0.264	0.144	0.018	-5.583	0.064	-132.763	0.013	0.170	0.203	-0.127	0.024
S.E.	(0.204)	(0.123)	(0.152)	(0.221)	(0.219)	(0.472)	(0.325)	(7.605)	(0.078)	(227.168)	(0.042)	(0.134)	(0.188)	(0.307)	(0.070)
R.I. <i>p</i> -value	[0.471]	[0.648]	[0.207]	[0.241]	[0.258]	[0.708]	[0.957]	[0.686]	[0.332]	[0.555]	[0.848]	[0.167]	[0.413]	[0.652]	[0.683]
Parent and Principal	0.048	0.205	0.279	0.063	0.239	0.292	0.212	9.487	0.124	-83.142	-0.001	0.263	0.285	-0.102	0.075
S.E.	(0.176)	(0.148)	(0.182)	(0.217)	(0.197)	(0.400)	(0.257)	(10.948)	(0.057)	(233.325)	(0.046)	(0.122)	(0.210)	(0.330)	(0.043)
R.I. <i>p</i> -value	[0.868]	[0.349]	[0.240]	[0.804]	[0.309]	[0.433]	[0.499]	[0.434]	[0.055]*	[0.710]	[0.975]	[0.023]**	[0.253]	[0.730]	[0.201]
Interaction	0.045	-0.033	-0.111	-0.210	0.111	0.359	0.363	25.729	-0.007	35.737	0.084	-0.095	0.107	0.069	0.019
S.E.	(0.438)	(0.285)	(0.341)	(0.385)	(0.317)	(0.547)	(0.454)	(19.521)	(0.091)	(363.079)	(0.056)	(0.169)	(0.394)	(0.325)	(0.106)
R.I. <i>p</i> -value	[0.887]	[0.914]	[0.763]	[0.562]	[0.743]	[0.537]	[0.426]	[0.104]	[0.937]	[0.904]	[0.310]	[0.579]	[0.752]	[0.871]	[0.833]
N (# students)	798	797	799	793	797	795	792	739	781	385	786	689	363	734	773
C (# strat. cells)	26	26	26	26	26	26	26	26	26	25	26	26	25	26	26
Adjusted R ²	0.200	0.178	0.208	0.141	0.143	0.156	0.227	0.014	0.045	0.170	0.177	0.159	0.304	0.122	0.083
Control-group values															
Mean	0.886	0.219	0.876	0.731	0.771	2.805	3.030	16.073	0.766	553.218	0.923	0.541	0.000	8.016	0.875
SD	1.609	0.769	1.571	1.406	1.561	2.095	2.151	48.393	0.425	604.364	0.267	0.500	1.000	1.553	0.332

Notes: Sample is parents of students with a valid endline literacy assessment, enrolled in one of the 103 study schools at the beginning of the 2025/26 academic year, who completed the endline parent survey; observation counts vary across columns with item non-response (don't-know and refused responses are treated as missing). The Engagement Index in column (13) is the first principal component of the outcomes in columns (1)–(11), with component weights estimated in the control group, each item standardized by its control-group mean and standard deviation, and the index standardized by its control-group distribution; the index is computed for parents answering every component item, so its sample is limited by non-response to the expenses question. Columns (14) and (15), separated by the column space, are not components of the index. Because no baseline exists for these outcomes, regressions control for a full set of age-category-by-gender interactions, parent education indicators, and stratification cell fixed effects, but no lagged dependent variable. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses. Randomization-inference *p*-values in brackets permute the four-arm assignment within the original stratification cells, clustered by school (1,000 draws); all statistics use the same permutation draws. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table 7
Program Impact on Student Reports of Parent Engagement

	(1) Family Reads Days [0-7]	(2) Family Tells a Story Days [0-7]	(3) Practices Reading Days [0-7]	(4) Talks About School Days [0-7]	(5) Regular Study Time 0/1	(6) Student Index SDs	(7) Sibling Supervises 0/1
Parent	0.538	0.288	0.554	0.753	0.132	0.556	-0.118
S.E.	(0.419)	(0.457)	(0.467)	(0.390)	(0.116)	(0.339)	(0.067)
R.I. <i>p</i> -value	[0.160]	[0.488]	[0.214]	[0.055]*	[0.380]	[0.086]*	[0.170]
Principal	0.537	0.459	0.673	0.653	0.060	0.386	-0.173
S.E.	(0.364)	(0.406)	(0.356)	(0.330)	(0.107)	(0.286)	(0.078)
R.I. <i>p</i> -value	[0.176]	[0.241]	[0.127]	[0.116]	[0.693]	[0.239]	[0.053]*
Parent and Principal	0.467	0.036	0.491	0.404	0.314	0.436	-0.046
S.E.	(0.321)	(0.292)	(0.322)	(0.320)	(0.090)	(0.216)	(0.080)
R.I. <i>p</i> -value	[0.285]	[0.943]	[0.321]	[0.338]	[0.028]**	[0.223]	[0.603]
Interaction	-0.607	-0.711	-0.736	-1.003	0.121	-0.505	0.244
S.E.	(0.505)	(0.536)	(0.585)	(0.490)	(0.150)	(0.414)	(0.105)
R.I. <i>p</i> -value	[0.283]	[0.235]	[0.235]	[0.075]*	[0.587]	[0.288]	[0.052]*
N (# students)	1,053	1,022	1,041	1,058	1,203	905	1,191
C (# strat. cells)	24	24	24	24	24	24	24
Adjusted R ²	0.091	0.083	0.136	0.124	0.201	0.178	0.144
Control-group values							
Mean	1.623	1.307	1.588	1.673	0.399	0.000	0.479
SD	1.633	1.529	1.615	1.670	0.490	1.000	0.500

Notes: Sample is students with a valid endline literacy assessment, enrolled in one of the 103 study schools at the beginning of the 2025/26 academic year, who completed the endline student survey; observation counts vary across columns with item non-response. The Student Index in column (6) is the first principal component of the outcomes in columns (1)–(5), with component weights estimated in the control group, each item standardized by its control-group mean and standard deviation, and the index standardized by its control-group distribution; the index is computed for students answering every component item. Column (7), separated by the column space, is not a component of the index: an indicator that the student names a sibling as the person who typically helps with schoolwork, among students naming any arrangement (students reporting no one are included; missing responses are not). Because no baseline exists for these outcomes, regressions control for a full set of age-category-by-gender interactions, parent education indicators, and stratification cell fixed effects, but no lagged dependent variable. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses. Randomization-inference *p*-values in brackets permute the four-arm assignment within the original stratification cells, clustered by school (1,000 draws); all statistics use the same permutation draws. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table 8
Treatment Effects on the Treated for Parent Intervention

	(1) Three-Item Index	(2) Three-Item Index	(3) Two-Item Index	(4) Two-Item Index	(5) Attended Either Meeting	(6) Attended Either Meeting
	ORF (SDs)	Reading Comp. (SDs)	ORF (SDs)	Reading Comp. (SDs)	ORF (SDs)	Reading Comp. (SDs)
Panel A: Treatment Effects on Compliance						
Parent Assignment	0.365***	0.367***	0.391***	0.387***	0.575***	0.571***
S.E.	(0.040)	(0.040)	(0.044)	(0.041)	(0.057)	(0.053)
Effective F (Olea-Pflueger)	85.10	83.97	80.40	91.30	102.77	113.76
Adjusted R ²	0.471	0.470	0.429	0.433	0.463	0.465
Panel B: Effect on Reading Scores (OLS)						
Compliance	0.282*	0.517*	0.360***	0.282***	0.234***	0.211***
S.E.	(0.153)	(0.252)	(0.107)	(0.094)	(0.067)	(0.061)
N (# students)	396	396	890	890	890	890
Panel C: Effect on Reading Scores (2SLS)						
Compliance	1.473***	0.996***	0.846**	0.410	0.576**	0.278
S.E.	(0.555)	(0.338)	(0.411)	(0.298)	(0.278)	(0.201)
N (# students)	689	689	1,698	1,698	1,698	1,698
C (# strat. cells)	26	26	26	26	26	26

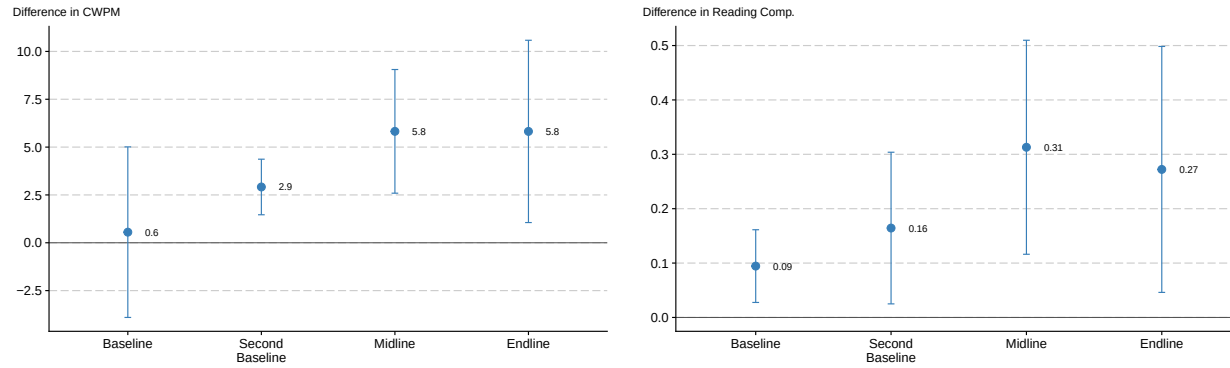
Notes: Sample in Panels A and C is all students with a valid endline literacy assessment, enrolled in one of the 103 study schools at the beginning of the 2025/26 academic year; Panel B restricts to students assigned to the parent program (parent-only and combined arms). Columns (1) and (2) further restrict to students whose surveyed parent answered the workbook question, which drives their smaller samples. Compliance measures are in natural units from zero to one: columns (1)–(2) use the mean of recorded attendance at each parent meeting and having the program workbook at home; columns (3)–(4) use the mean of the two attendance indicators; columns (5)–(6) use an indicator for recorded attendance at at least one meeting. Attendance comes from digitized meeting sign-in sheets, with unrecorded students counted as not attending; workbook receipt comes from the endline parent survey. Panel A reports the effect of assignment to the parent program on each compliance measure; the effective F statistic is the Olea–Pflueger robust first-stage statistic. Panel B reports OLS regressions of reading scores on compliance among treated students, which do not account for selection into compliance. Panel C instruments compliance with assignment to the parent program. Outcomes in Panels B and C are in standard deviations of the control-group distribution, so coefficients give the implied effect of moving from no compliance to full compliance. All regressions control for the baseline and second-baseline values of the dependent variable with missing values zeroed out and indicated, a full set of age-category-by-gender interactions, parent education indicators, stratification cell fixed effects, and indicators for principal-program assignment. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses. Randomization inference is not reported in this table: permuted assignments generate first stages near zero, so the randomization distribution of the two-stage estimator is uninformative; the corresponding randomization-based tests are the reduced-form intent-to-treat p -values in Table 4 (0.045 for fluency and 0.215 for comprehension). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table 9
Program Impact on Principal and Teacher Practices

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	Observations of First-Grade Teachers			Teacher Practices				Teacher Agreement (1-5)		Principal Reports		
	App-Logged Observations Count	Principal- Reported Times/Term	Teacher- Reported Times/Term	Marks Workbooks Mult./Week 0/1	Adapts Most Lessons 0/1	One-on-One Parent Meetings 0/1	Ever Stressed by Feedback 0/1	Head Teacher Helps Instruction 1-5	Parents Have Knowledge 1-5	Parents Have Knowledge 1-5	Discusses ORF with Parents 0/1	One-on-One Parent Meetings 0/1
Parent	1.399	-0.602	1.832	-0.088	0.128	0.303	-0.092	0.049	0.649	0.841	-0.058	0.149
S.E.	(1.714)	(2.035)	(2.077)	(0.125)	(0.199)	(0.170)	(0.186)	(0.274)	(0.397)	(0.583)	(0.199)	(0.192)
R.I. <i>p</i> -value	[0.513]	[0.708]	[0.251]	[0.398]	[0.482]	[0.052]*	[0.654]	[0.868]	[0.049]**	[0.121]	[0.749]	[0.371]
Principal	3.363	-0.377	0.269	-0.231	-0.235	-0.025	0.342	-0.490	0.091	-0.554	-0.077	-0.124
S.E.	(2.829)	(1.859)	(2.048)	(0.136)	(0.208)	(0.242)	(0.277)	(0.417)	(0.489)	(0.645)	(0.215)	(0.229)
R.I. <i>p</i> -value	[0.104]	[0.821]	[0.874]	[0.013]**	[0.191]	[0.884]	[0.083]*	[0.094]*	[0.774]	[0.294]	[0.664]	[0.466]
Parent and Principal	3.133	-0.098	2.713	-0.080	-0.226	0.295	0.065	0.115	0.351	1.115	0.442	0.024
S.E.	(2.078)	(2.134)	(1.503)	(0.079)	(0.207)	(0.150)	(0.261)	(0.260)	(0.376)	(0.468)	(0.213)	(0.271)
R.I. <i>p</i> -value	[0.127]	[0.953]	[0.116]	[0.353]	[0.206]	[0.055]*	[0.751]	[0.726]	[0.357]	[0.033]**	[0.012]**	[0.900]
Interaction	-1.630	0.881	0.613	0.240	-0.119	0.016	-0.185	0.556	-0.389	0.828	0.577	-0.001
S.E.	(3.271)	(2.254)	(2.931)	(0.162)	(0.281)	(0.284)	(0.322)	(0.533)	(0.478)	(0.743)	(0.322)	(0.247)
R.I. <i>p</i> -value	[0.640]	[0.672]	[0.780]	[0.091]*	[0.638]	[0.945]	[0.502]	[0.149]	[0.434]	[0.276]	[0.030]**	[0.998]
N (# schools)	90	60	68	70	67	66	70	67	70	60	59	59
C (# strat. cells)	26	16	25	25	25	25	25	25	25	16	16	16
Adjusted R ²	0.230	-0.050	0.110	0.406	0.160	0.169	-0.217	0.179	-0.126	0.158	0.103	0.049
Control-group values												
Mean	5.458	5.133	5.895	0.900	0.650	0.650	0.350	4.158	3.550	3.133	0.267	0.667
SD	5.710	4.228	4.326	0.308	0.489	0.489	0.489	0.602	0.887	1.302	0.458	0.488

Notes: Sample is the study schools with students in the endline analysis sample, with one observation per school; coverage varies across columns with the data source. Column (1) counts lesson observations of first-grade teachers logged on the SmartCoach application after the January training. Columns (2)–(9) come from the endline teacher survey and columns (10)–(12) from the endline principal survey; no principal surveys exist for private schools, so the principal-survey columns (and column (2), the principal’s report of observation frequency) cover government schools only. Reported observation frequencies in columns (2) and (3) are converted to times per term (never = 0; once per term = 1; two to three times per term = 2.5; two to three times per month = 7.5; weekly or more = 12). Regressions are at the school level and control for stratification cell fixed effects only. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses. Randomization-inference *p*-values in brackets permute the four-arm assignment within the original stratification cells (1,000 draws); all statistics use the same permutation draws. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Figure 5
Parent Meeting Attenders vs Non-Attenders



Panel A: Reading Fluency

Panel B: Reading Comprehension

Note: Sample is children in schools assigned to the parent intervention with digitized attendance records from both parent meetings (26 schools; 604 children). Attendance indicates that a parent signed in at either of the two meetings; attendance is not randomly assigned. Each point reports the coefficient on an attendance indicator from a separate regression of that wave's score on age-by-gender cells and stratification-cell fixed effects; waves after baseline additionally control for the baseline score and an indicator for a missing baseline assessment. Numbers beside each point report the estimate; whiskers show 95% confidence intervals based on standard errors clustered by stratification cell. Sample sizes vary by wave with assessment coverage (413–545 children). Parent meetings began December 15, 2025; second-baseline assessments were collected between late November and early January, largely before the first meeting, so the significant differences at and before the second baseline reflect selection into attendance rather than effects of the meetings.

5 Conclusion

Education systems worldwide are scaling up education interventions that have proven to greatly increase literacy. Many of these target classroom instruction, such as Kenya’s TU-SOME structured pedagogy program, which has reached over five million students (Piper et al., 2018b). Many others target other inputs to education, such as India’s Rocket Learning initiative, which successfully engaged over 150,000 parents in their children’s education using behavioral nudges (Goyal et al., 2024). The counterfactual of a world filled with prolific, poorly run education programs is disappearing. These programs alone will not be enough: most of Africa is over 3 standard deviations (PIRLS, 2021) behind high-income countries. Multiple programs will have to work together improving classrooms, and engaging parents and communities, and improving management at schools, and reducing the cost of schooling, and all work together. Policymakers have scarce evidence on how these approaches fit together. Unexpected interactions between program components are an important cause of “voltage drop” when scaling (List, 2026).

To begin collecting this evidence, I study two interventions that target inputs to learning outside the classroom and implement them in schools already receiving a successful structured pedagogy program. The parent intervention produces strong learning gains as an add-on to structured pedagogy. It achieves these gains by leveraging untapped complementarities with the structured pedagogy materials. By sending home the literacy workbooks used in class, the parent intervention increases the productivity of learning time at home, which allows it to function even when siblings rather than parents are a child’s main learning supervisor. The principal intervention produces large reading comprehension losses. I attribute these learning losses to the training’s focus on documenting coaching sessions, shifting those sessions towards rigid, formal institutions that incentivize principals to detrimentally micromanage their teachers. The losses prey on negative complementarities with the structured pedagogy program. Less strict adherence to the lesson plans allowed teachers to adapt their lessons to better target their instruction to their students’ needs.

Unexpected behavioral responses are likely a key factor in education programs having unintended or unexpectedly small effects. I show that parents increased their engagement to

offset some of the principal intervention’s learning losses. This is consistent with education production being concave in parents’ investment. While in this case the household response was beneficial, if concavity holds everywhere, this behavioral response would lead parents to decrease investment if the principal intervention had increased learning. This crowd-out effect has been documented when school inputs improve (Das et al., 2013; Pop-Eleches and Urquiola, 2013). Behavioral responses may thus act as an automatic stabilizer for education: buffering students from harmful interventions but preventing them from realizing the full gains of beneficial ones.

Further research should continue studying education programs with the appropriate counterfactual: a successful education system rather than the old status quo. A/B testing has recently gained prominence in education (Angrist, Cullen, and Magat, 2025) as an effective way to achieve this goal. Continuous experimentation and improvement allow education systems to keep improving and closing the gap with rich countries. Even extremely successful programs can benefit from this approach. Nyqvist and Guariso (2026) show that community meetings can augment structured pedagogy, while Duflo, Kiessel, and Lucas (2020) tests multiple models of Teaching at the Right Level in Ghana and find there are still large gains to be found. Factorial experiments offer another path to estimating program interactions, although these take an extremely large sample size to achieve reasonable statistical power (Gelman, 2018). A key limitation of my study is that I am only able to bound the interaction between my two interventions. More research with larger sample sizes is needed to further understand program interactions.

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A Attrition & Balance

Table A1
Realized Design

		Parent Program	
		<i>No</i>	<i>Yes</i>
Principal Training	<i>No</i>	24 schools	23 schools
		(465 students)	(431 students)
	<i>Yes</i>	23 schools	21 schools
		(343 students)	(459 students)

Notes: 91 schools had endline data. The 103 schools were randomly assigned to one of the four main study arms as shown in the table. 5 private schools attrited before baseline treatment. 8 private schools did not run end-of-year assessments. (See [Table 1](#) for the assigned number of schools per arm)

Table A2
Attrition Table

	(1)	(2)	(3)	(4)
Parent	-0.009	-0.041	-0.038	0.110*
S.E.	(0.073)	(0.078)	(0.077)	(0.059)
Principal	0.010	-0.017	-0.013	0.074
S.E.	(0.081)	(0.074)	(0.073)	(0.082)
Parent and Principal	-0.072	-0.082	-0.084	0.035
S.E.	(0.072)	(0.077)	(0.076)	(0.036)
Male $\times$ Parent				-0.068
S.E.				(0.056)
Male $\times$ Principal				-0.101
S.E.				(0.075)
Male $\times$ Parent and Principal				-0.048
S.E.				(0.058)
Age 7 $\times$ Parent				0.043
S.E.				(0.059)
Age 7 $\times$ Principal				-0.088
S.E.				(0.094)
Age 7 $\times$ Parent and Principal				-0.105
S.E.				(0.069)
Age 8 $\times$ Parent				-0.045
S.E.				(0.089)
Age 8 $\times$ Principal				0.071
S.E.				(0.102)
Age 8 $\times$ Parent and Principal				0.009
S.E.				(0.081)
Missing Age $\times$ Parent				-0.225*
S.E.				(0.122)
Missing Age $\times$ Principal				-0.032
S.E.				(0.136)
Missing Age $\times$ Parent and Principal				-0.127
S.E.				(0.125)
Joint F-test	0.99	0.56	0.62	1.37
R.I. p -value	[0.511]	[0.749]	[0.725]	[0.939]
Stratification FE	No	Yes	Yes	Yes
Baseline controls	No	No	Yes	Yes
Observations	2,089	2,089	2,089	2,089
Clusters	26	26	26	26

Notes: Sample is all the students who were enrolled in one of the 103 study schools at the beginning of the 2025/2026 academic year. Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis (). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table A3
Post Attrition Balance Table – Students

	(1) Control Mean (SD)	(2) Parent Diff.	(3) Principal Diff.	(4) Parent and Principal Diff.	(5) Obs.
Male	0.497 (0.501)	-0.029 (0.614)	0.063 (0.438)	-0.028 (0.806)	1,698
Student Age (Years)	6.595 (0.723)	0.066 (0.619)	0.252* (0.056)	-0.085 (0.607)	796
Missing Baseline Assessment	0.155 (0.362)	0.023 (0.656)	-0.031 (0.270)	0.107** (0.027)	1,698
Baseline Letter Sounds	5.427 (2.404)	0.123 (0.331)	-0.017 (0.890)	-0.195 (0.213)	1,326
Baseline Decodable Words	2.987 (1.983)	-0.260 (0.416)	0.217 (0.439)	-0.141 (0.463)	1,326
Baseline Sight Words	4.384 (2.338)	0.090 (0.753)	-0.198 (0.508)	0.010 (0.962)	896
Baseline ORF	11.738 (12.021)	-3.158 (0.177)	-4.111* (0.066)	-0.385 (0.894)	717
Baseline Reading Comp.	0.600 (0.725)	-0.185 (0.285)	-0.133 (0.407)	-0.258 (0.108)	532
Missing Second Baseline Assessment	0.460 (0.499)	-0.025 (0.802)	-0.007 (0.949)	-0.097 (0.359)	1,698
Second Baseline ORF	18.570 (14.290)	1.882 (0.583)	-0.692 (0.883)	-1.923 (0.539)	897
Joint F-stat		12.26	3.07	5.09	
R.I. p -value		[0.185]	[0.833]	[0.527]	

Notes: Sample is all the students who present at endline data collection. Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis (). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table A4
Post Attrition Balance Table – Teachers

	(1) Control Mean (SD)	(2) Parent Diff.	(3) Principal Diff.	(4) Parent and Principal Diff.	(5) Obs.
Female	0.900 (0.308)	0.068 (0.434)	-0.008 (0.954)	0.062 (0.444)	67
Age (Years)	37.222 (10.138)	2.224 (0.528)	-2.341 (0.322)	-0.670 (0.864)	66
Years Teaching	11.399 (8.653)	-0.364 (0.907)	-0.971 (0.722)	-2.227 (0.513)	69
Years at Current School	4.377 (5.408)	0.865 (0.606)	0.393 (0.834)	1.077 (0.618)	67
Years Teaching Current Level	2.985 (4.159)	-0.101 (0.966)	0.670 (0.630)	-0.984 (0.585)	67
Joint F-stat		16.13	105.20	2.98	
R.I. p -value		[0.403]	[0.164]	[0.789]	

Notes: Sample is all the teachers who were teaching first-grade in one of the 103 study schools at endline. Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis (). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table A5
Post Attrition Balance Table – Principals

	(1) Control Mean (SD)	(2) Parent Diff.	(3) Principal Diff.	(4) Parent and Principal Diff.	(5) Obs.
Years Since Started Teaching	17.000 (8.494)	2.205 (0.588)	1.432 (0.628)	6.180 (0.145)	59
Years as School Leader	5.467 (4.998)	2.936 (0.342)	0.727 (0.790)	4.886 (0.161)	57
Years Leading Current School	4.867 (5.069)	-0.161 (0.940)	-0.136 (0.950)	1.753 (0.415)	57
Head Teacher (vs. Other Leader)	0.067 (0.258)	-0.023 (0.854)	0.048 (0.710)	0.292 (0.121)	60
Currently Teaching a Class	0.867 (0.352)	0.091 (0.542)	0.018 (0.906)	-0.247 (0.208)	60
Master's Degree or Higher	0.071 (0.267)	0.033 (0.831)	0.019 (0.883)	0.400** (0.047)	59
Lesson Observations Before Training	16.067 (14.058)	3.227 (0.610)	2.021 (0.735)	1.774 (0.811)	60
Joint F-stat		614.53	1.59	49.30	
R.I. <i>p</i> -value		[0.039]	[0.874]	[0.188]	
Joint F-stat (excl. missingness indicators)		83.89	14.12	6.24	
R.I. <i>p</i> -value		[0.021]	[0.095]	[0.212]	

Notes: Sample is all the principals who were working in one of the 103 study schools at endline. Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis (.). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table A6
Post Attrition Balance Table – Schools

	(1) Control Mean (SD)	(2) Parent Diff.	(3) Principal Diff.	(4) Parent and Principal Diff.	(5) Obs.
Students on Rolls	22.542 (16.832)	-1.703 (0.769)	-7.184 (0.213)	-0.163 (0.983)	91
Share Assessed at Baseline	0.764 (0.362)	-0.074 (0.229)	-0.000 (0.993)	-0.085* (0.069)	91
Share Male	0.478 (0.208)	-0.004 (0.952)	0.064 (0.380)	0.005 (0.959)	91
School Mean Baseline ORF	4.105 (4.807)	-0.738 (0.569)	-0.632 (0.653)	-0.905 (0.591)	91
School Mean Baseline Reading Comp.	0.161 (0.269)	-0.046 (0.451)	-0.019 (0.811)	-0.100 (0.192)	91
Joint F-stat		2.11	0.44	3.17	
R.I. p -value		[0.288]	[0.746]	[0.129]	

Notes: Sample is all the schools still operating at endline Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis ().
*: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table A7
Post Attrition Balance Table – Parents

	(1) Control Mean (SD)	(2) Parent Diff.	(3) Principal Diff.	(4) Parent and Principal Diff.	(5) Obs.
Parent Survey Completed	0.434 (0.496)	0.157 (0.106)	0.060 (0.418)	0.041 (0.673)	1,698
Parent Age (Years)	39.877 (10.673)	-2.968* (0.086)	-2.298* (0.093)	-1.279 (0.425)	638
Parent Reads English	0.648 (0.479)	0.058 (0.318)	0.009 (0.912)	-0.056 (0.488)	784
Parent Has No Schooling	0.183 (0.388)	-0.085* (0.070)	-0.103* (0.100)	0.002 (0.974)	803
Parent Completed SHS or Higher	0.267 (0.444)	0.052 (0.199)	0.071 (0.280)	-0.008 (0.884)	803
Parent Lives with Child	0.915 (0.280)	0.030 (0.384)	-0.002 (0.957)	0.054 (0.151)	798
Parent Has Older Children	0.812 (0.392)	-0.042 (0.421)	-0.031 (0.654)	0.073 (0.128)	803
Household Assets [0-7]	3.317 (1.561)	-0.536** (0.017)	-0.454* (0.083)	-0.265 (0.365)	803
Joint F-stat		6.91	169.16	1.13	
R.I. p -value		[0.296]	[0.056]	[0.898]	

Notes: Sample is all the parents who were found and surveyed. Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis (). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table A8
Parent Survey Attrition Table

	(1)	(2)	(3)	(4)
Parent	-0.109	-0.157	-0.160*	-0.110
S.E.	(0.106)	(0.094)	(0.093)	(0.137)
Principal	-0.017	-0.060	-0.059	0.011
S.E.	(0.076)	(0.073)	(0.072)	(0.112)
Parent and Principal	-0.027	-0.041	-0.049	0.078
S.E.	(0.100)	(0.095)	(0.094)	(0.159)
Male $\times$ Parent				0.037
S.E.				(0.048)
Male $\times$ Principal				0.144*
S.E.				(0.071)
Male $\times$ Parent and Principal				-0.066
S.E.				(0.074)
Age 7 $\times$ Parent				0.074
S.E.				(0.125)
Age 7 $\times$ Principal				0.004
S.E.				(0.150)
Age 7 $\times$ Parent and Principal				-0.104
S.E.				(0.123)
Age 8 $\times$ Parent				-0.103
S.E.				(0.176)
Age 8 $\times$ Principal				0.037
S.E.				(0.188)
Age 8 $\times$ Parent and Principal				-0.041
S.E.				(0.118)
Missing Age $\times$ Parent				-0.108
S.E.				(0.128)
Missing Age $\times$ Principal				-0.319**
S.E.				(0.146)
Missing Age $\times$ Parent and Principal				-0.140
S.E.				(0.207)
Joint F-test	0.39	0.94	0.99	3.70
R.I. p -value	[0.812]	[0.534]	[0.517]	[0.632]
Stratification FE	No	Yes	Yes	Yes
Baseline controls	No	No	Yes	Yes
Observations	1,698	1,698	1,698	1,698
Clusters	26	26	26	26

Notes: Sample is all the parents who were found and surveyed. Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis (.). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table A9
Parent Attendee vs Non-Attendee Characteristics

	(1) Did Not Attend Mean (SD)	(2) Attended Diff.	(3) Obs.
Panel A: Student Characteristics			
Male	0.539 (0.500)	-0.085* (0.086)	604
Student Age (Years)	6.724 (0.770)	-0.125 (0.179)	365
Missing Baseline Assessment	0.186 (0.390)	-0.136*** (0.001)	604
Baseline Reading Index	-0.000 (1.000)	0.245* (0.063)	545
Joint F-stat		15.77 [0.000]	
Panel B: Parent Characteristics			
Parent Survey Completed	0.392 (0.489)	0.165** (0.029)	604
Parent Age (Years)	40.852 (11.784)	-2.398* (0.066)	259
Parent Reads English	0.545 (0.501)	0.043 (0.422)	303
Parent Has No Schooling	0.250 (0.436)	-0.050 (0.312)	311
Parent Completed SHS or Higher	0.263 (0.443)	0.047 (0.432)	311
Parent Lives with Child	0.950 (0.219)	0.048* (0.082)	309
Parent Has Older Children	0.825 (0.382)	-0.038 (0.325)	311
Household Assets [0-7]	2.763 (1.512)	0.066 (0.696)	311
Joint F-stat		5.20 [0.003]	

Notes: Sample is children in parent-intervention schools with digitized attendance records from both parent meetings. "Attended" indicates a parent signed in at either meeting; attendance is not randomly assigned, so differences are descriptive. Column 1 reports means among non-attenders with standard deviations in parentheses. Column 2 reports the coefficient on attendance from a regression of each characteristic on an attendance indicator and stratification-cell fixed effects; p-values based on standard errors clustered by stratification cell appear in parentheses. Panel B characteristics are observed only for the parent-survey completers in each group. Joint F-statistics come from regressing attendance on all covariates in the panel (with missingness indicators) and stratification-cell fixed effects. Joint F-statistic based on [Kerwin, Rostom, and Sterck \(2024\)](#). Heteroskedasticity-robust standard errors, clustered by stratification cell, in parenthesis (.). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

B All EGRA Subtests

During the final wave of EGRA data collection, my intent was for all students to receive the full battery of subtests. To achieve this, ITG updated SmartCoach but only a subset of teachers installed the new update. Because of this, only teachers at 41 schools (29 government and 12 private) ran the full EGRA covering 491 students. This was a non-random sample of schools and in particular the principal intervention only cell only six schools ran the full EGRA. So, these estimates should be interpreted with caution. [Table B1](#) presents estimates for that sample from all subtests.

Table B1

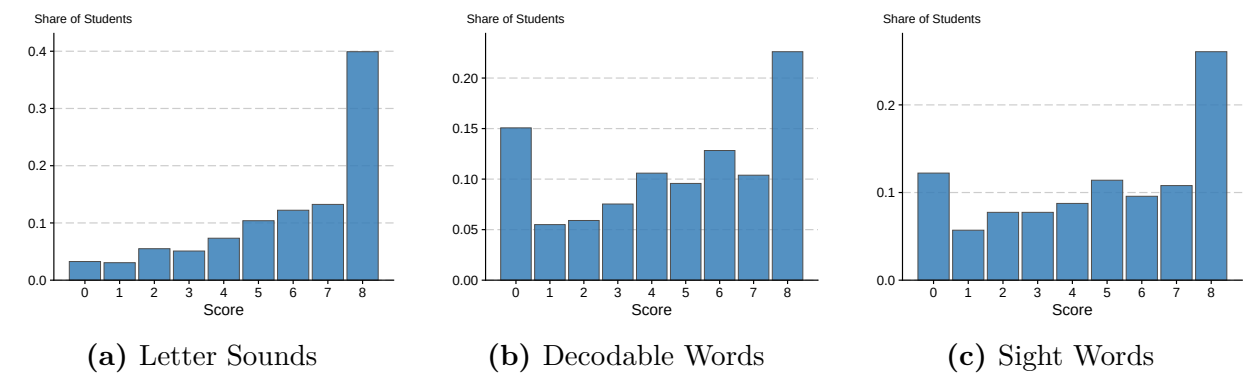
	(1) Letter Sounds	(2)	(3) Decodable Words	(4)	(5) Sight Words	(6)	(7) Oral Reading Fluency	(8)	(9) Reading Comprehension	(10)	(11) Reading Index	(12)
	Score [0-8]	SDs	Score [0-8]	SDs	Score [0-8]	SDs	CWPM	SDs	Score [0-3]	SDs	PCA	SDs
Parent and Principal	-1.412	-0.645	-0.662	-0.237	-1.077	-0.405	-2.394	-0.080	0.012	0.010	-0.107	-0.107
S.E.	(1.159)	(0.529)	(0.940)	(0.337)	(0.977)	(0.367)	(9.559)	(0.321)	(0.310)	(0.265)	(0.362)	(0.362)
R.I. <i>p</i> -value	[0.405]		[0.556]		[0.425]		[0.781]		[0.966]		[0.735]	
Parent	-1.408	-0.643	-0.583	-0.209	-1.176	-0.443	4.424	0.148	0.066	0.057	-0.208	-0.208
S.E.	(0.927)	(0.423)	(0.848)	(0.304)	(0.813)	(0.306)	(7.055)	(0.237)	(0.205)	(0.175)	(0.261)	(0.261)
R.I. <i>p</i> -value	[0.213]		[0.561]		[0.219]		[0.602]		[0.819]		[0.541]	
Principal	-1.080	-0.493	-0.773	-0.277	-0.979	-0.368	-2.314	-0.078	-0.502	-0.429	-0.339	-0.339
S.E.	(0.900)	(0.411)	(0.755)	(0.270)	(0.807)	(0.303)	(5.745)	(0.193)	(0.218)	(0.186)	(0.270)	(0.270)
R.I. <i>p</i> -value	[0.265]		[0.344]		[0.260]		[0.808]		[0.143]		[0.195]	
N (# students)	491		491		491		491		491		491	
C (# strat. cells)	23		23		23		23		23		23	
Adjusted R ²	0.282		0.213		0.348		0.365		0.324		0.442	
Control-group values												
Mean	6.066		4.628		5.033		35.355		1.281		0.000	
SD	2.190		2.793		2.658		29.817		1.171		1.000	
Pooled = Parent + Principal, R.I. <i>p</i> -value	[0.358]		[0.563]		[0.389]		[0.759]		[0.353]		[0.356]	

Notes: Sample is all the students who were enrolled in one of the 103 study schools at the beginning of the 2025/2026 academic year who received the full battery of EGRA subtests at endline. Treatment effects estimated using a linear regression of the outcome on an indicator for each treatment arm, an indicator for gender, continuous age, their interactions, baseline values of the outcome, and stratification cell indicators. Heteroskedasticity-robust standard errors, clustered by stratification cell and calculated by bootstrap with 1,000 simulations reported in parenthesis (). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

There is an additional issue with the estimates for letter sounds, decodable word reading, and sight word reading. Each of these subtests was designed to be shorter than the standard EGRA to not impose too much time on teachers. Instead of the usual block of 100 letters and 50 words students can attempt to read, these tests included just eight letters and words each which introduced ceiling effects. Ceiling effects occur when a large portion of students achieve the maximum score on a testing instrument. [Figure B1](#) shows the distribution of outcomes those skills. Each of them has a high share of outcomes bunched at the top of

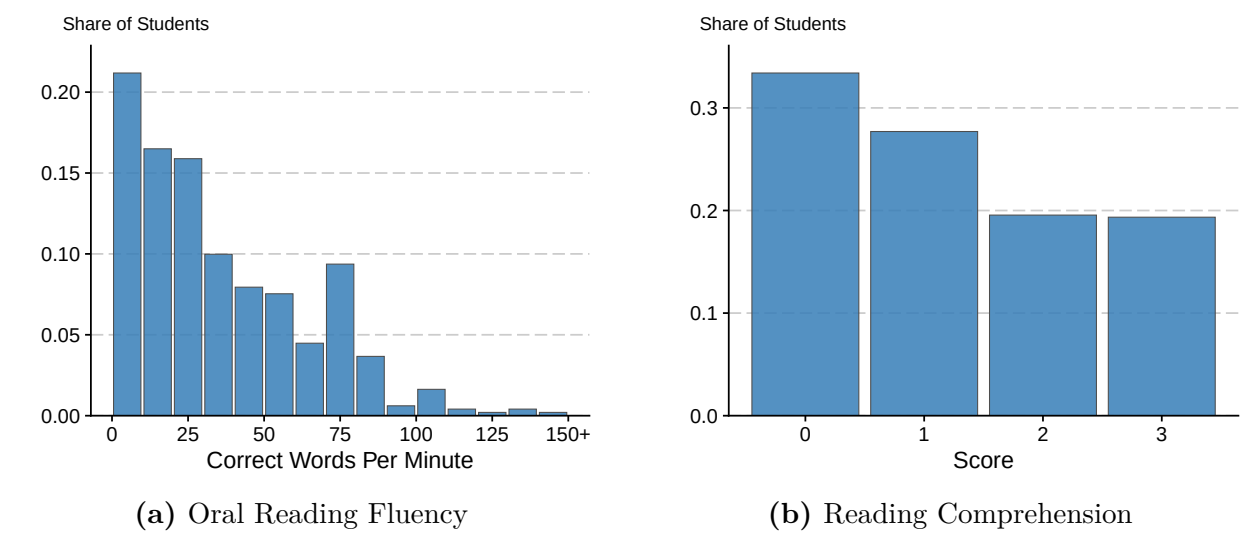
the distribution. In contrast, [Figure B2](#) shows the distribution for the reading fluency and comprehension where there is a mostly smooth spread of scores. Ceiling effects can make estimates of treatment effects unreliable by masking variation for students with higher skills which can even flip the sign of estimated effects ([Wang et al., 2008](#)).

Figure B1
Endline Basic-Skills Score Distributions



Notes: Sample is the 491 students assessed with the full EGRA instrument at endline. Each subtest is scored out of 8.

Figure B2
Endline Fluency and Comprehension Distributions



Notes: Sample is the 491 students assessed with the full EGRA instrument at endline. Oral reading fluency is grouped in 10-word bins, with the top bin collecting scores above 150.

C Robustness

Table C1
Treatment Robustness

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
		Oral Reading Fluency (SDs)				Reading Comprehension (SDs)		
Parent	0.331	0.297	0.246	0.314	0.159	0.182	0.205	0.158
S.E.	(0.170)	(0.186)	(0.164)	(0.163)	(0.121)	(0.142)	(0.123)	(0.118)
R.I. p -value	[0.045]**	[0.090]*	[0.147]	[0.053]*	[0.215]	[0.203]	[0.112]	[0.212]
Principal	-0.106	-0.133	-0.154	-0.118	-0.262	-0.194	-0.208	-0.263
S.E.	(0.109)	(0.134)	(0.099)	(0.109)	(0.101)	(0.122)	(0.102)	(0.100)
R.I. p -value	[0.528]	[0.466]	[0.349]	[0.486]	[0.048]**	[0.175]	[0.114]	[0.047]**
Parent and Principal	0.193	0.134	0.194	0.186	0.093	0.029	0.115	0.093
S.E.	(0.144)	(0.184)	(0.130)	(0.144)	(0.107)	(0.140)	(0.095)	(0.107)
R.I. p -value	[0.219]	[0.471]	[0.231]	[0.234]	[0.468]	[0.828]	[0.377]	[0.460]
Interaction	-0.032	-0.030	0.102	-0.010	0.196	0.042	0.117	0.198
95% CI	[-0.43, 0.37]	[-0.46, 0.40]	[-0.28, 0.49]	[-0.40, 0.38]	[-0.11, 0.50]	[-0.33, 0.41]	[-0.22, 0.45]	[-0.10, 0.50]
R.I. p -value	[0.887]	[0.898]	[0.663]	[0.964]	[0.316]	[0.828]	[0.525]	[0.305]
N (# students)	1,698	1,698	1,698	1,698	1,698	1,698	1,698	1,698
C (# clusters)	26	26	26	24	26	26	26	24
Adjusted R ²	0.300	0.152	0.216	0.291	0.334	0.144	0.267	0.328
Baseline Score Controls	Yes	No	Yes	Yes	Yes	No	Yes	Yes
Demographic Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Strat. Cell FE	Original	Original	None	Adjusted [†]	Original	Original	None	Adjusted [†]

Notes: Sample is all the students who were enrolled in one of the 103 study schools at the beginning of the 2025/2026 academic year. Column 1 is the pre-registered specification from Table 1. Column 2 drops the baseline and second-baseline score controls and their missingness indicators. Column 3 drops the stratification cell fixed effects. Column 4 reassigns schools in stratification cells left without variation in treatment status by whole-school attrition to the cell one lower ([†] cells 25 and 26, each containing only their control school after attrition, are combined into cell 24). Demographic controls (age-category-by-gender interactions and parent education) are included in all columns. Heteroskedasticity-robust standard errors, clustered by stratification cell (adjusted cells in column 4), in parentheses. Randomization-inference p -values, clustered by school and stratified by the original stratification cells, in brackets. *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table C2
Lee Bounds for Reading Outcomes

	(1) Oral Reading Fluency (SDs)	(2) Reading Comprehension (SDs)
Parent		
Upper Bound	0.373**	0.167
S.E.	(0.185)	(0.136)
Lower Bound	0.274	0.121
S.E.	(0.202)	(0.135)
Trimming Share	0.012	0.012
Principal		
Upper Bound	-0.032	-0.180
S.E.	(0.117)	(0.128)
Lower Bound	-0.067	-0.239*
S.E.	(0.136)	(0.133)
Trimming Share	0.019	0.019
Parent and Principal		
Upper Bound	0.200	0.107
S.E.	(0.162)	(0.139)
Lower Bound	0.029	-0.036
S.E.	(0.158)	(0.135)
Trimming Share	0.079	0.079
Interaction		
Upper Bound	-0.007	0.224
S.E.	(0.299)	(0.257)
Lower Bound	-0.312	-0.023
S.E.	(0.239)	(0.229)

Notes: Sample is all the students who were enrolled in one of the 103 study schools at the beginning of the 2025/2026 academic year. Treatment effects estimated using a linear regression of the outcome on an indicator for each treatment arm, an indicator for gender, continuous age, their interactions, baseline values of the outcome, and stratification cell indicators. Heteroskedasticity-robust standard errors, clustered by stratification cell and calculated by bootstrap with 1,000 simulations reported in parenthesis (). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$.

Table C3
One-Sided Hypothesis Tests for Main Results

	(1) Oral Reading Fluency	(2) SDs	(3) Reading Comprehension	(4) SDs
	CWPM		Score [0-3]	
Parent and Principal	5.383	0.193	0.102	0.093
S.E.	(4.028)	(0.144)	(0.118)	(0.107)
R.I. <i>p</i> -value (two-sided)	[0.219]		[0.468]	
Parent	9.224	0.331	0.176	0.159
S.E.	(4.746)	(0.170)	(0.133)	(0.121)
R.I. <i>p</i> -value (one-sided)	[0.019]**		[0.109]	
R.I. <i>p</i> -value (two-sided)	[0.045]		[0.215]	
Principal	-2.959	-0.106	-0.290	-0.262
S.E.	(3.037)	(0.109)	(0.111)	(0.101)
R.I. <i>p</i> -value (one-sided)	[0.730]		[0.972]	
R.I. <i>p</i> -value (two-sided)	[0.528]		[0.048]	
N (# students)	1,698		1,698	
C (# strat. cells)	26		26	
Adjusted R ²	0.300		0.334	
Control-group values				
Mean	30.320		1.127	
SD	27.874		1.106	
Pooled = Parent + Principal, R.I. <i>p</i> -value (two-sided)	[0.887]		[0.316]	

Notes: Specification and sample are identical to Table 1. Inference follows the pre-analysis plan: the Parent and Principal main effects are secondary hypotheses tested one-sided against $H_1: \beta > 0$ (randomization-inference p = share of permuted estimates at least as large as the observed estimate), rejecting at the 5% level only when the estimate is positive; two-sided p -values are reported alongside for transparency. Estimated negative effects with two-sided $p < 0.05$ are descriptive findings outside the formal hypothesis tests. The interaction test in the final row is the pre-registered primary literacy hypothesis and is two-sided, as is the total effect of receiving both programs. Randomization-inference p -values permute the four-arm assignment within the original stratification cells, clustered by stratification cell. Heteroskedasticity-robust standard errors, clustered by stratification cell, in parentheses. *: $p < 0.1$; **: $p < 0.05$; *** : $p < 0.01$ (one-sided rows starred on the one-sided p).

D Other Tables & Figures

Table D1
Conditional Main Effects

	(1) Oral Reading Fluency	(2) Reading Fluency	(3) Reading Comprehension	(4) Reading Comprehension
	CWPM	SDs	Score [0-3]	SDs
Parent, in Principal Schools	8.342	0.299	0.393	0.355
S.E.	(3.093)	(0.111)	(0.081)	(0.074)
R.I. <i>p</i> -value	[0.066]*		[0.007]***	
Principal, in Parent Schools	-3.841	-0.138	-0.073	-0.066
S.E.	(4.722)	(0.169)	(0.144)	(0.130)
R.I. <i>p</i> -value	[0.386]		[0.601]	
Interaction	-0.882	-0.032	0.217	0.196
S.E.	(5.455)	(0.196)	(0.165)	(0.149)
R.I. <i>p</i> -value	[0.887]		[0.316]	
N (# students)	1,698		1,698	
C (# strat. cells)	26		26	
Adjusted R ²	0.300		0.334	
Control-group values				
Mean	30.320		1.127	
SD	27.874		1.106	

Notes: Sample and specification are identical to Table 4; all statistics are linear combinations of that regression's arm coefficients. The first row reports the effect of the parent program among schools assigned the principal program (schools receiving both programs versus schools receiving only the principal program); the second row reports the effect of the principal program among schools assigned the parent program. The Interaction row equals the difference between each conditional effect and its unconditional counterpart in Table 4. Standard errors are computed from the full covariance matrix of the arm coefficients, clustered by stratification cell, in parentheses. Randomization-inference *p*-values in brackets permute the four-arm assignment within the original stratification cells, clustered by school (1,000 draws, identical to Table 1). *: $p < 0.1$; **: $p < 0.05$; ***: $p < 0.01$ (one-sided rows starred on the one-sided *p*).

Table D2
School Activities at Home Supervisor

	(1)	(2)	(3)	(4)
	Control	Parent	Principal	Parent and Principal
Mother	0.306	0.404	0.444	0.362
Father	0.082	0.098	0.152	0.138
Sibling	0.479	0.330	0.292	0.422
Grandparent	0.020	0.044	0.047	0.030
No One	0.093	0.118	0.064	0.046
Other	0.020	0.007	0.000	0.003

Notes: *: $p < 0.1$; **: $p < 0.05$; *** : $p < 0.01$ (one-sided rows starred on the one-sided p).

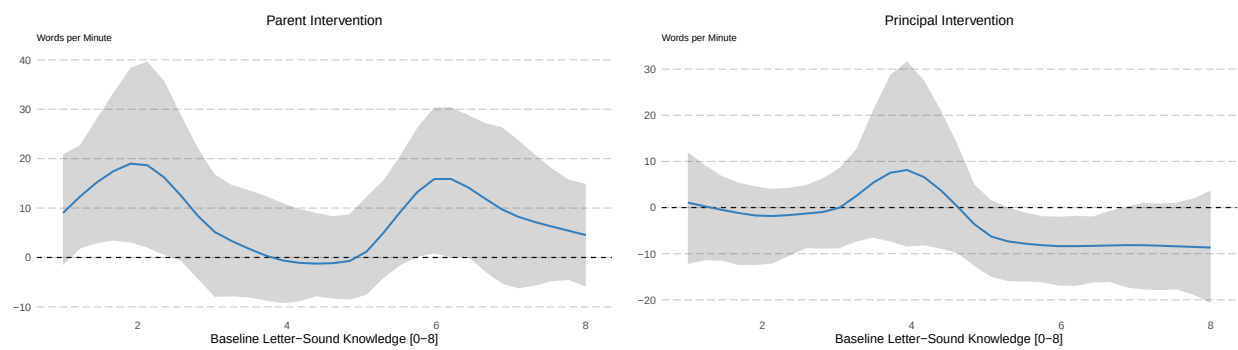
E Baseline Heterogeneity

In this section, I show baseline heterogeneous treatment effects. I do this in two ways. First, I use the baseline score on letter sound knowledge. Second I use the baseline value of the same test as the stratification. [Figure E1](#) and [Figure E2](#) show the estimates. Of these, my preferred estimate stratifies on letter sound knowledge, but both should be interpreted with caution.

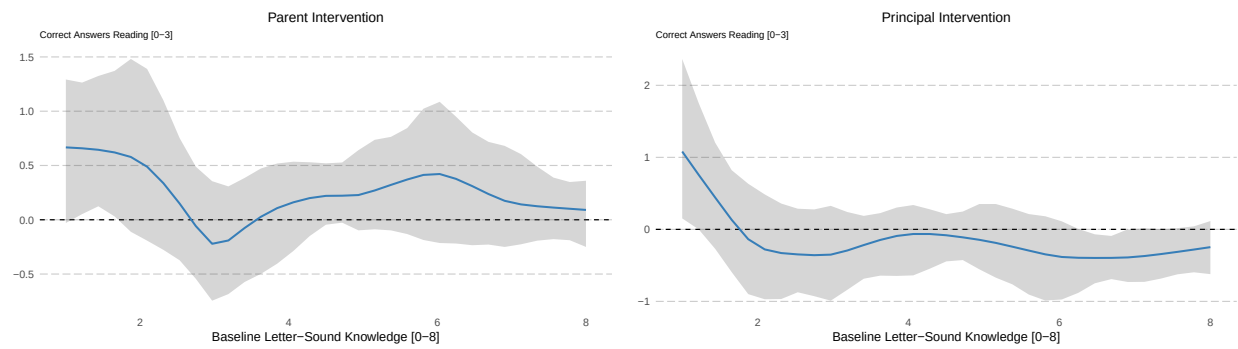
I prefer the estimates that stratify with letter sound knowledge because of limited coverage of advanced skill assessment at baseline. The baseline EGRA collection had limited students reach the advanced subtests of reading fluency and comprehension despite high overall assessment rates. This is because of standard EGRA stopping rules. The version of the EGRA I used at baseline had five subtests which were in order: letter sounds, decodable words, sight words, oral reading fluency, and reading comprehension. If a student fails to cross a (low) bar of correct answers on a given subtest, the assessment ends and they do not move on to any other subtests. For example if a student could not read a single word on the oral reading fluency subtest, they would not attempt reading comprehension. This is done to not demoralize students. Because of the stopping rule, all students took the letter sound knowledge subtest, but only 54% reached oral reading fluency and only 40% reached reading fluency. In [Figure E2](#) I use the full sample by imputing a score of zero for any student who took the baseline EGRA but did not reach the relevant subtest. These imputations follow the spirit of the EGRA stopping rule but will not be accurate for all students who may have reading skills that do not match the EGRAs progression.

Both estimates should be interpreted with caution. Letter sound knowledge correlates with oral reading fluency but noisily and so has limited explanatory power. It is also a basic reading skill that noisily proxies true reading ability. Thus, stratifying by letter sound knowledge reveals little of differential treatment effects for students at different baseline reading abilities. For the self-stratified figures, while the underlying baseline skill better reveals reading ability heterogeneity, the majority of observations are imputed zeros. So, a meaningful amount of variation is lost.

Figure E1
Treatment Effects Stratified by Baseline Letter Sounds Score

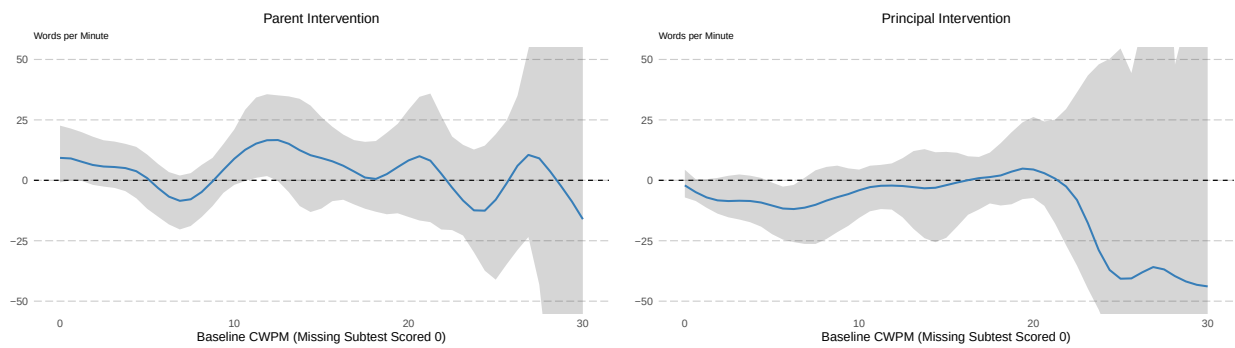


Panel A: Oral Reading Fluency

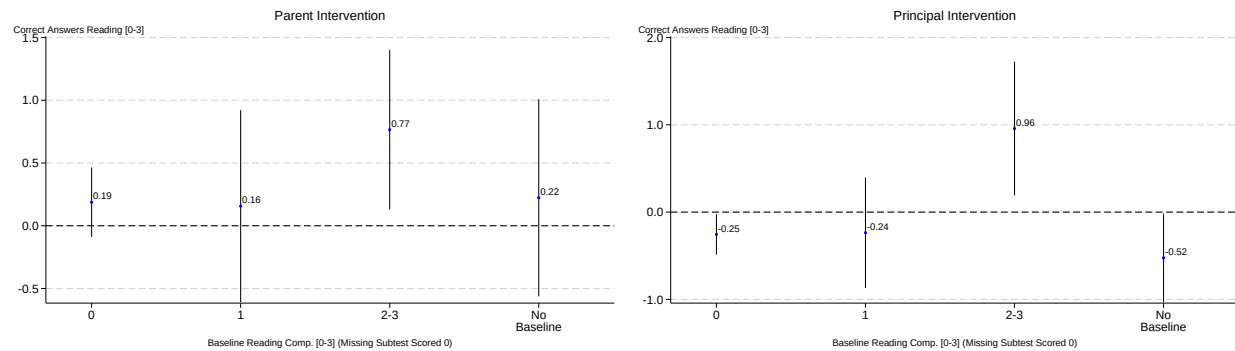


Panel B: Reading Comprehension

Figure E2
Treatment Effects Stratified by Baseline Value of Own Test



Panel A: Oral Reading Fluency



Panel B: Reading Comprehension

F Parent Intervention Materials

Figure F1
Home Reading Activity

Home Reading Activity

Term 2

Spend 5 minutes doing this set of activities with your child every day at home to help improve their reading.

To get started, scan the QR code:

Watch literacy expert, Regina Mensah, demonstrate the sound that each letter in the English alphabet makes.



1. Letter Sounds

Practise saying the sounds letters make:

u	c	k	w	j	v	y	z
x	qu	ck	nk	ng	sh	ch	th

2. Sight Words

Practise reading these high frequency words:

but	up	play	get	be	him
we	will	jump	come	have	are
you	yes	her	how	what	box
here	was	went	has	then	or
put	one	they	eat	with	why

Figure F2
Literacy Report Card

Literacy Progress Report Card

Term 2

Name:

Class:

Date:

Oral Reading Fluency

Prosody

CWPM

Low

Med

High

By the end of Term 2, we hope learners are able to correctly read **20 or more words** in a minute (CWPM).

High prosody is reading with expression, using pauses and stresses in the right places.

Week 9 Assessment

Oral Language

Phonics

Reading

Writing

/4

/10

/3

/3

Class Teacher Signature:

To learn more about what this report card means, turn to [page 130](#).

Head Teacher signature:

Date signed:

Turn to [Page 158](#) to start the Home Reading Activity

BS1 | Term 2

161

Report Card

68

Figure F3
Parent Meeting Facilitator Guide Page 1

Term 2 Parent Meeting Facilitator's Guide

Week of 23-27 March 2026

🕒 60 min

At a Glance

Attendees: Basic 1 Parents

Facilitators: Headteacher or Assistant
Headteacher and Basic 1 Teacher

Purpose: To encourage parent involvement and
strengthen the partnership between home and
school in supporting children's reading.

Materials

- ☐ Basic 1, Term 2 Inspiring Reading workbooks
(full class set)
- ☐ Class roster
- ☐ Parent sign-in sheet
- ☐ Board and chalk/marker

To Do Before the Meeting

- ☐ Finish your Term 2 ORFs using SmartCoach.
- ☐ Fill-out each learner's Report Card in the
back of the workbook on page 161.
- ☐ Check that quiz scores are filled in on page
162.

End of the Meeting

- ☐ Parents should take their child's workbook
home at the end of the meeting.
- ☐ By the last day of the term, send home
workbooks for any child who did not have a
parent attend the meeting.

💡 Distribute your learners' workbooks to their parents.

1 Welcome and Introduction

🕒 10 min

Headteacher Leads

Good morning, parents.

We are very happy you are here today. Thank
you for coming. We know you are busy, and
we appreciate your time.

When you attend meetings like this, it sends a
strong message to your child that education
matters.

Before we begin, we want to thank you.

Since our last meeting, we have seen an
improvement in the way our learners read.
Some are reading more words. Some are
attempting difficult words. Some are more
confident in class. Some are correcting
themselves while reading.

This improvement comes from two places:

- The hard work of our teachers.
- The support you are giving at home.

When school and home work together,
children improve faster.

Learning to read takes time. Some children
learn quickly. Some learn slowly. Both are
normal. What matters is progress.

Today we will:

1. Reflect on what happened after our last
meeting.
2. Look at your child's progress.
3. Understand what progress really looks
like.
4. Review Words Per Minute and the Report
Card.
5. Practise again how to support reading at
home.

We are one team in your child's success.

Figure F4
Parent Meeting Facilitator Guide Page 2

2
Recap of Term 1
⌚ 15 min

Basic 1 Teacher Leads

Last term at the parent meeting, we talked about your role in helping your child become a reader. We said even 10–15 minutes a few times a week can make a big difference.

We introduced the Inspiring Reading workbook and explained that it is not only for school, it is also a tool for home practice.

We showed you the Literacy Progress Report Card at the back of the workbook. This report card shows your child's reading level and progress.

We explained Words Per Minute. This is how many correct words your child reads in one minute.

We also talked about fluency. Fluency is important because when children read smoothly, they understand better. If they struggle to read each word, they cannot focus on the meaning.

Reflection Discussion

After our last meeting, how did you practise with your child?

> Pause and allow parents to respond.

Prompts to get parent responses:

- Did your child read to you?
- Did you read together?
- Did you talk about the pictures?
- Did you practise the sight words?
- Did you try timing them?

Thank you for sharing. Even small efforts help.

3
Looking at Progress This Term
⌚ 10 min

Please open your child's workbook.

Look at this early story from the beginning of Term 2 on page 5.

Notice the short words and simple sentences.

Now look at a recent story on page 123.

Notice the longer words and more complex sentences.

Your child has learned more letter sounds this term. When children know more sounds, they can read more words.

If a child knows only a few sounds, they can read only a few words.

The number of words they can read in one minute should increase over time.

Progress is not about comparing your child to others.

Progress is about comparing your child to themselves.

Ghana has set the goal that a child can correctly read 45 words per minute by the end of Basic 2.

We are building toward that goal step by step.

Figure F5
Parent Meeting Facilitator Guide Page 3

4
What Progress Looks Like
⌚ 15 min

Write these letters on the board:

chop hat shop

Progress can be seen in many different ways. Sometimes it even looks like “mistakes.” Let me show you.

Pretend I am now a Basic 1 learner trying to read these words on the board.

Demonstrate each:

- Chip (Attempting to read a word, but getting it wrong.)
- H-a-t (Breaking a word into sounds.)
- S, h, o, p (Saying letter names.)
- Ship, chip (Correcting themselves.)

All of the examples I just demonstrated are positive and show learning. We can praise these actions. Over time, they will get the words “correct.”

If your child corrects himself/herself, that is progress.

If they try to spell a word, that is progress.

If they explain a story to you, that is progress.

Learning to read is like building a house - one step at a time.

Reflection Discussion

Prompts for parents to share:

- What have you noticed at home that shows your child is learning to read?

> Allow parents to share. Encourage and affirm responses.

5
Understanding the Report Card and Fluency
⌚ 10 min

Turn to the back of the workbook on page 161.

Here it says, CWPM (Correct Words Per Minute.)(point to this)

Words Per Minute means how many correct words your child reads in one minute. The goal is that at the end of Basic 1, Term 2 learners can read 20 words per minute.

By the end of Grade 2, the goal is 45 words per minute.

Smooth reading helps understanding. When reading becomes automatic, children can focus on meaning.

How can you help improve fluency?

- Listen to your child read aloud.
- Ask them to read the same story again.
- Encourage smoother reading.
- Ask simple questions about the story.

Reading a story again builds speed and confidence.

**Note - Each term, the words in the story on the assessment get a little harder. This can cause some learners’ scores not to change as much.*

Figure F6
Parent Meeting Facilitator Guide Page 4

6 Two Languages Are a Strength ⌚ 15 min

Many of our children speak Fante at home and learn English at school.

Some parents worry:

- “If I speak Fante at home, will it confuse my child?”
- “Should I stop speaking Fante so my child can learn English faster?”
- “Do I need to speak English well to help my child?”

We want to reassure you. Speaking two languages is not a problem. It is a strength.

When a child learns two languages, their brain becomes stronger at thinking, remembering, and understanding. Language builds language.

If your child can tell a full story in Fante — with a beginning, middle, and end — that storytelling skill helps them understand stories in English.

If your child knows many words in Fante, they are building vocabulary skills. When they learn new English words, their brain already understands how words work.

For example:

If your child can explain, “Yesterday we went to the market to buy fish,” in Fante, they are learning how to organise thoughts, use full sentences and explain events clearly.

Those same thinking skills help them explain things in English.

A strong first language, supports a strong second language!

You do not need to stop speaking your home language. Instead, you can use both languages to help your child grow.

Here are simple things you can do..

Suggestions for parents:

- Look at a picture in the workbook and ask your child to describe it in Fante.
- Then ask them to describe it again in English.
- Tell a story in Fante and ask them to retell it in English.
- Ask them to explain what they learned in school in either language.

If you feel unsure about helping in English, you can:

Suggestions for parents:

- Ask an older sibling to listen to the reading.
- Ask another family member to help.
- Practise together — your child reading English, and you discussing the meaning in Fante.

Understanding is more important than perfect pronunciation. Children who feel proud of their home language are more confident learners.

We want our learners to be strong in both Fante and English.

Figure F7
Parent Meeting Facilitator Guide Page 5

7
Demonstration
⌚ 15 min

Now we are going to practise together.

I am going to pretend that I am the parent, and this is my child at home.

I will show you how to help in simple ways.

Let's open the workbook to the story on page 129.

First, I look at the picture and say:

"Tell me about this picture."

"What do you see?"

"What is happening here?"

I listen carefully. I allow the child to speak. I do not rush them.

Then I move to reading.

"Read the story to me."

Allow the child to read.

If they make a mistake, sometimes I help, or sometimes I ignore the mistake and focus on the fact that they are trying.

I encourage them by saying: "Good job."

After they finish, I say: "Now read it again, but this time try to read it more smoothly."

Reading again helps children become more confident and more fluent.

Then I ask:

"What happened in the story?"

"Who was in the story?"

"Why did that happen?"

Asking questions helps children understand what they read, not just say the words.

8
Parent-Child Reading Session
⌚ 10 min

In this session, the child will read to their parent out of their workbook. You should prepare your learners to do this by explaining and practicing with them before the meeting.

Pre-meeting preparation with children:

- Prep children ahead of time.
- Put a simple task list on the board.
- Let the child lead.
- Keep teacher prompts minimal and respectful.
- End with a structured goal-sharing moment.

Now we are going to give the your children the opportunity to lead.

Parents, today your child will show you what they have been learning.

We have prepared them ahead of time so they know what to do.

Please sit next to your child and open their workbook.

Write on the board

Today I Will Show My Parent:

- A decodable story
- A Friday quiz

Children, you may begin!

Part 1: Decodable Story

- Children will choose a recent decodable story and read it to their parent.
- Teachers walk around quietly and support if needed.
- After the first read, ask children, read it one more time. This time, try to read smoothly.

Let's move to the next part. Children you may talk about your Friday Quiz!

Figure F8
Parent Meeting Facilitator Guide Page 6

Part 2: Friday Quiz

Children turn to a recent Friday quiz. They should explain to their parent:

- What the quiz was about.
- What they found easy or challenging.

Now we are going to end with something important.

Children, tell your parents something you are doing well with in school.

> Pause and allow sharing.

Now, you will share a goal. What do you want to be in the future?

> Pause

Great work sharing, children.

9 Closing

⌚ 10 min

Headteacher Leads

Parents, before we end, I want to thank you for coming. Thank you for helping your children at home. Small steps each week turn into big results over time.

When you sit beside your child and listen to them read, something powerful happens.

They feel: "My parent believes in me." "My reading matters." "I can do this."

Confidence builds. They gain more skills. And have more success.

You are shaping your child's future.

A strong reader understands better, stays in school longer, and has more opportunities in life.

Teachers will continue working hard. But we need you.

Even 10–15 minutes at home makes a difference. And, you can come to your child's teacher and me to ask questions about school and their learning.

Encourage your child. Praise their effort. Keep going.

Your words matter. Your time matters. Your presence matters. And we are proud of you.

Let us continue working together. Thank you very much.

G Principal Intervention Materials

Figure G1
Principal Training Facilitator Example 1

As you watch the video, fill the blanks below:

Module 1	Introduction to Lead _____ And a Structured Pedagogy Approach
Module 2	Research-Informed _____ Teaching Techniques, Assessments & Remediation
Module 3	Research-Informed _____ Teaching Techniques, Assessments & Remediation
Module 4	_____ FLN In Your _____ How to launch, run and sustain your Lead FLN Program

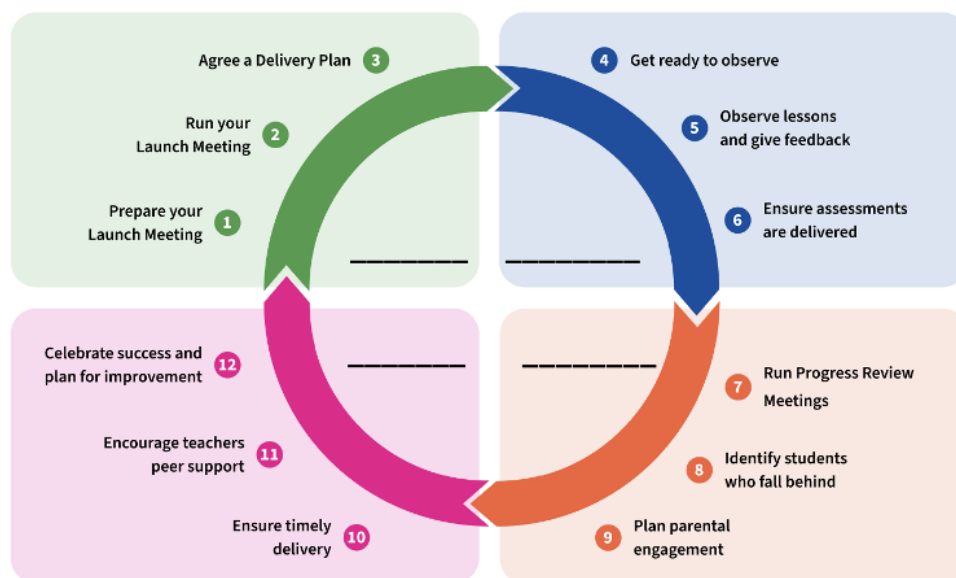
Figure G2
Principal Training Facilitator Example 2

Literacy Techniques
4 Key Areas

			
1 Teach New Vocabulary	4 Build _____ Awareness	7 Blending Activities	10 Teach Writing Skills & Conventions
2 Interactive Read _____	5 Letter-Sound Drills	8 Teach Spelling & _____	11 Shared Writing Practice
3 Teach Language Structures	6 Teach New Concepts	9 Structured Reading Practice	12 _____ Writing

Figure G3
Principal Training Facilitator Example 3

Implementation Cycle



**Fill
in the
blanks**